Exact Quantum Dynamics on the Minimal (13,233) Holographic Substrate

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Abstract

The paper provides a minimal exact numeric realisation of the holographic principle. The objectless, phase sector of quantum dynamics is exhibited complete on the smallest non-trivial finite holographic substrate determined by a pair of finite fields: the Carrier $\mathbb{F}_{233}$ and the Subject $\mathbb{F}_{13}$. The total state of physical space-time is the wave function, realised as the register on the holographic encoding of the observer's sky. Schrödinger unitary evolution is the rigid rotation of that encoding. The Dirac flow is a gauge deformation forced by the finiteness itself: the Carrier is a torsor, a field only once the frame of reference is selected by the Subject, and no bounded observer embeds without deficit. The deficit carries a finite rate, one Carrier tick per chronon, which no infinite substrate can host; its two faces, apsidal advance and clock dilation, support the gravitational realisation: gravity as the finite-substrate correction to the rotation, standing with its strong-field falsifier, the double cover supplying exactly the factor that relates the faces. The observable sky is the past light cone, realized event by event as a section of a finite Hopf fibration. Every structural theorem is proven at general p, drawn at the instance in a live interactive laboratory, and machine-verified in integer arithmetic end to end; the continuum enters only as labelled charts. The paper is the laboratory's certificate.

Keywords: finite fields, quantum dynamics, Schrödinger equation, Dirac flow, quantum gravity, model existence proof, finite ring cosmology, Hopf fibration, holographic principle, exact integer verification

1 Introduction

The state-of-the-art quantum theory has no complete worked instance. Its formalism lives in Hilbert space over the continuum, where even a finite-dimensional system carries continuous amplitudes and no claim can be checked to the last integer, and every interpretational debate inherits that slack: the measurement problem, the origin of complex amplitudes, and the status of the wave function are argued over structures that no computation exhausts.

Finite and discrete reconstructions attack this slack from four directions, each from an assumed input. The oldest line puts quantum mechanics on finite arithmetic while keeping its postulates: Schwinger's unitary operator bases opened finite-dimensional quantum kinematics [1]; Wootters formulated finite-state quantum mechanics on a discrete Wigner function [2], refined to the discrete phase space over a finite field [3]; Galois-indexed Hilbert spaces organise these structures systematically [4]; modal quantum theory [5] and Galois-field quantum mechanics [6] replace the complex amplitudes themselves and study which phenomena survive; and Lev's finite-mathematics programme rebuilds the whole mathematical foundation over rings $\mathbb{Z}/p$, recovering standard theory as a degenerate limit [7], with the finitist reading of analysis itself argued in [8]. The assumed input, throughout: the Hilbert-space postulates — states, observables, the Born rule — imposed on the new arithmetic, with the observer outside the formalism. An information-theoretic argument reaches the same finitism from the numbers themselves: a generic real number carries infinite information, so a physics of finite information must discard the real line, and classical determinism dissolves with it [9]; already the choice of mathematical language, classical or intuitionist, fixes what time can mean [10]. Here the assumed input is the inherited dynamics: the equations stay continuum, only their number field is trimmed.

A second line posits a deterministic or computational substrate beneath the quantum states: Wheeler's it-from-bit places information at the bottom [11], the cellular-automaton interpretation posits an exact deterministic evolution beneath Hilbert space [12], and the computational-universe programme a discrete update rule [13]. The assumed input is the rule itself, and quantum structure is recovered by superimposed templates. A third line makes spacetime discrete while importing matter and the quantum postulates: spin networks build space from combinatorial data [14], causal sets from a partial order [15], energetic causal sets from unique events carrying energy and momentum, time taken as fundamental and irreversible [16], loop gravity from connection variables [17, 18], causal dynamical triangulations from a regularised path integral [19]. The fourth line is the strongest external evidence that the physical state count is finite: the black-hole entropy bound [20], dimensional reduction [21], the holographic principle [22], and the de Sitter horizon entropy [23] — read, in those programmes, as constraints on continuum theories rather than as the substrate itself. Alongside these, the relational and epistemic readings of quantum mechanics make states observer-relative [24] or reconstruct quantum phenomenology from a knowledge balance [25]: relational registration, without a finite substrate beneath it. The natural-philosophy case that the universe is singular and time is real [26] states two premises the present construction enforces arithmetically: one totality, and the observer's time as its own drive.

In every one of these programmes the finite structure is imposed on quantum mechanics from outside, or the quantum structure is imposed on the finite substrate from outside; none exhibits a complete instance in which the dynamics, the observables, and the observer are all finite and all derived. A structural line completes the survey: Noether's identity of symmetry and conservation [27], and Henkin's model existence, a consistent theory building its own model from its own terms [28]. The claim of the present construction is that these lines are one: the pair $(13, 233)$ is an explicit model, exhibited by construction from its own terms; its symmetries are its conserved content; its sky is its holographic chart; its registered data are its bits. Nothing here discretises a continuum theory.

The finite ring continuum (FRC) framework inverts the direction: the finite substrate is the single structure, and quantum dynamics is derived from it. A bounded observer, the *Subject*, is a prime field $\mathbb{F}_\mathsf{p}$ with $\mathsf{p} = 4\kappa + 1$; the totality it inhabits, the *Carrier*, is a field $\mathbb{F}_\Omega$ with $\Omega = 4S + 1$; the observer's time is its own drive $x \mapsto \mathsf{g}x$. The algebra, the geometry, and the dynamics of this substrate are constructed in [29–32]; the present paper needs no result beyond those it states and proves below.

The picture the instance draws is one sentence long. The total state of physical space-time is the wave function, and the wave function is the register: its holographic encoding on the observer's sky. *Quantum evolution is the rigid rotation of that encoding*: the Schrödinger unitary evolution is the rotation itself, exactly (§5). The Dirac flow dressing the rotation is a gauge deformation, and it exists because the Carrier is finite: a torsor with no origin, unit, or clock of its own, a field only once the Subject frames it, and one in which no bounded Subject embeds without deficit. The deficit has a rate, one Carrier tick per chronon; on an infinite substrate the tick vanishes and the deformation with it, so the flow is the finite signature of the substrate (§9.1). Its two faces, apsidal advance and clock dilation, support the gravitational realisation: the deformation as the finite-substrate correction to the rotation, standing with its strong-field falsifier (Statement 11).

This paper exhibits the complete Objectless phase sector of quantum dynamics on the minimal non-trivial pair $(\mathsf{p}, \Omega) = (13, 233)$: $\kappa = 3$, $S = 58$, no Object. The contribution is fourfold.

1. **The laboratory, the core result** (§1.1, §3): the complete model drawn live on a public interactive page, every drawn element a stated law of this paper, the rendering itself machine-verified.

2. **The structural theorems, self-contained** (§4): the six load-bearing theorems of the quantum sector, stated and proven at general $\mathsf{p} = 4\kappa + 1$ — integration by quotient plus

cover, the spinor dichotomy, the Fourier-flip quarter, the precession law κ/S, the Hopf section, and the mass–energy channel $C_{\mathsf{p}-1} \cap C_{2(\mathsf{p}+1)} = Q_4$.

3. **The instance, complete** (§2.1, §5–§8): every theorem instantiated at $(13, 233)$ with displayed arithmetic — the tower identity $\mathsf{g} = 3 \cdot 5 = 2$, the quarter roots $\hbar = 144$ and $h = 89$ with $\hbar + h = \Omega$, the apsidal fraction $3/58$, the node covering $144 = 36 + 108$, the fold fusion of eight ray-ends into four horizon points, and the register reading $\psi_{\tau+1} = i\,(3\psi_\tau)$ of the Schrödinger step.
4. **The verification** (§10.2): six suites, 192 integer-exact checks, plus the five audits distributed with this paper (85 exact checks and 10 symbolic identities); every exact claim below names its witness.

Throughout, a claim tagged exact is an identity of finite arithmetic, checked by machine; continuum-valued quantities appear only as declared chart conventions, marked [approx]. The construction consumes no premise beyond the pair itself; in particular no physical constant, and no numeral of the physical capacity, enters anywhere.

1.1 The laboratory

The core result is drawn before it is derived. The complete model runs live at `https://frc-lab.gamma.earth/38-s13`: a single-file interactive laboratory presenting one state, at one scale, in three complementary projections; Figure 1 shows the production rendering of two of them. One toggle switches the Subject and Carrier views: the Carrier view shows the gauge, the common fiber flow; the Subject view shows the observable, the static retarded curl; their difference is the Dirac deformation of the picture above, live. Every state is exact and discrete, every drawn element is a stated law of this paper, and the rendering itself is machine-verified (`verify-render`, 20 checks, within the 192-check suite of §10.2). The rest of the paper is the laboratory's certificate: §3 details the page panel by panel, §4 proves the structures it draws, and §6–§8 derive the drawn laws row by row.

2 The substrate: a primer

Carrier, Subject, register. The Carrier is a finite field $\mathbb{F}_\Omega$, $\Omega = 4S+1$, carrying no distinguished origin, unit, generator, or clock: its reframings $x \mapsto ax + b$ are automorphisms of the affine line (torsor reframings, not field maps), and all marks below are frame data of an observer, never absolute structure. A Subject is a prime field $\mathbb{F}_{\mathsf{p}}$, $\mathsf{p} = 4\kappa + 1$, realised inside the Carrier; its *frame* is the tuple $(\tau; 0, 1, \mathsf{g})$: an origin, a unit, and the Subject's own drive generator. The *register* is the Subject's array of residues; registration is writing a value into it.

Registration, not embedding. No ring embedding exists between shells of different characteristic, and none is used: a Subject is registered against the Carrier through shared structure alone, the common class quotient ($\gcd(12, 232) = 4$ at the instance), the octant transport of Theorem 1, and the joint recurrence of the pair space. Cross-register readings are typed: a quantity of one shell combines with another's only through a declared transport, never by addition across moduli [33]. The transport is explicit: the class quotients are the capacity powers

$$\pi_{\mathsf{p}} : C_{\mathsf{p}-1} \to Q_4,\ x \mapsto x^\kappa, \qquad \pi_\Omega : C_{\Omega-1} \to C_4,\ y \mapsto y^S$$

(at the instance $x \mapsto x^3$ with kernel C_3 and $y \mapsto y^{58}$ with kernel C_{58}), identified through the orientation $\iota : i \mapsto \hbar$ (Proposition 7; A5). The registration object is the fibre product

$$R = \{(x, y) : \pi_{\mathsf{p}}(x) = \iota^{-1}\pi_\Omega(y)\}, \qquad |R| = \tfrac{(\mathsf{p}-1)(\Omega-1)}{4} = 696 :$$

the pair space itself, whose cardinality is the joint recurrence of §6 and whose exhaustion between (home, home) events is the closure of the frame leak (`check_s13.js`).

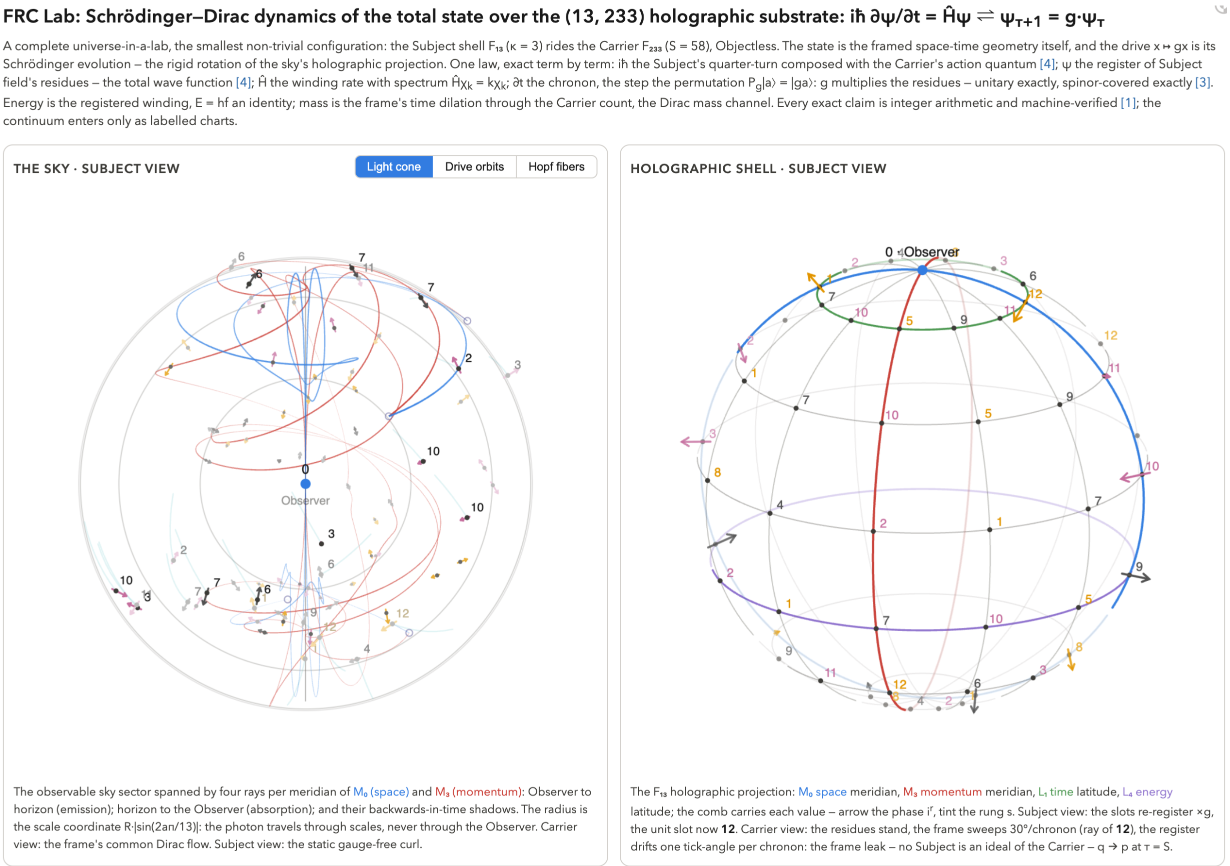


Figure 1: The laboratory (production rendering from `https://frc-lab.gamma.earth/38-s13`): the sky (left) and the holographic encoding shell (right), one state under the Subject/Carrier toggle; the sky's representation toggle (light cone / drive orbits / Hopf fibers) visible in the panel header. Every drawn law of §6–§8 is on this page; the rendering itself is verified by `verify-render` (20 checks).

Chronon, capacity, drive. The chronon τ is the observer's tick, the single application of the drive $x \mapsto \mathsf{g}x$. The capacity κ bounds resolution: $\mathsf{p} = 4\kappa + 1$, and a Subject resolves κ nested scales (§6). Time is not a Carrier datum: each Subject rides its own drive and assigns its own tick; what two shells share is arithmetic structure, never a clock.

The quarter and the fold. Since $\mathsf{p} \equiv 1 \pmod 4$, the drive cycle $C_{\mathsf{p}-1}$ contains a unique four-element subgroup $Q_4 = \{1, i, -1, -i\}$ with $i^2 = -1$. Every register value folds as $x = i^r \mathsf{g}^s$: the class r is the registrable phase, the rung s the outcome index; observation resolves cosets of the fold, never points within one. The fold clock $r = \tau \bmod 4$ is the quarter's trace in time.

The framed shell, drawn. Figure 2 draws the frame data at the paper's own Subject: the orbital 2-sphere and its framed-complex Euclidean chart, two readings of one shell. The derived constants close on the shell: $i = -\mathsf{g}^\kappa = 5$, $e = \mathsf{g}^i = 6$, $\pi = 2\kappa = 6$, with $2\pi \equiv -1$ and $\mathsf{g}^\pi \equiv -1$ (Proposition 7); every mark, the quarter-turn included, is a finite-field residue of the one shell.

Split and non-split tori; the spinor cover. Over $\mathbb{F}_\mathsf{p}$ the multiplicative group $C_{\mathsf{p}-1}$ is the split torus; the quadratic extension $\mathbb{F}_{\mathsf{p}^2}$ supplies the non-split (norm-one) torus $C_{\mathsf{p}+1}$, the finite boost circle, and its double cover $C_{2(\mathsf{p}+1)}$, the spinor cover. Mass sits on the non-split component (Realisation 8); the interplay of the two tori is the content of Theorem 6.

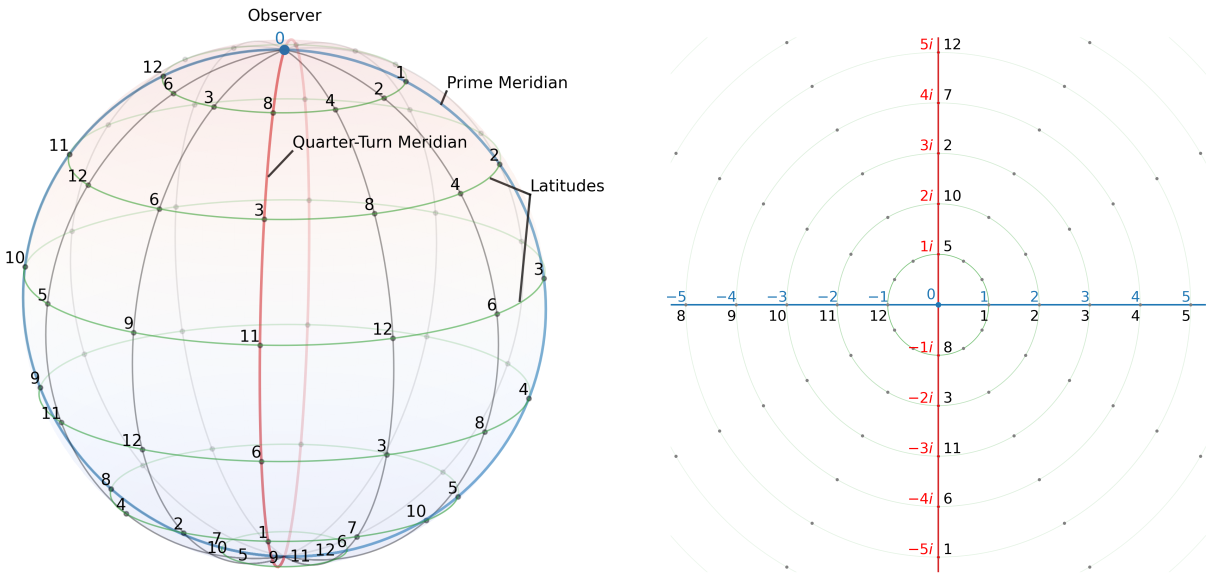


Figure 2: The framed shell $\mathbb{F}_{13}(\tau; 0, 1, \mathsf{g})$ of capacity $\kappa = 3$ [33]: chronon τ (the observer's now), position 0 (the here), scale 1, generator $\mathsf{g} = 2$, oriented quarter-turn $i = -\mathsf{g}^{\kappa} = 5$, exponential unit $e = \mathsf{g}^{i} = 6$, half-period $\pi = 2\kappa = 6$. Left: the orbital 2-sphere, the origin at the pole, the additive prime meridian (blue), the quarter-turn meridian (red), the drive latitudes (green). Right: the framed-complex Euclidean chart $a + b\,i$ over $\mathbb{F}_{13}$: the prime meridian the real axis, the quarter-turn meridian the imaginary axis, the latitudes norm circles. Every mark is a residue of the one shell.

Admissibility. An admissible pair satisfies $\mathsf{p} = 4\kappa + 1$ and $\Omega = 4S + 1$ prime with $S \equiv 2 \pmod 4$, $S \equiv 1 \pmod 3$, $\Omega \equiv 5 \pmod{12}$, and $\mathsf{p}^2 < \Omega$ (the Subject resolvable within the Carrier's coherence horizon); jointly, $\Omega \equiv 41 \pmod{48}$. The clauses' derivations live in the corpus [30, 32]; theorems below that consume a clause state it. Every occurrence of "minimal" in this paper is minimality under this predicate.

Multi-body registration. An unequal-cycle composite registers jointly on the lcm of its parts' periods, and each pair's conserved offset lives on the gcd cycle; dephasing over the product period is exact. The instance uses this once, in the sky chronon (§6).

The four domains, two registers. Figure 3 assembles the architecture: the two charts and their Fourier duals, the four representation domains, read in both registers, one row per register. The Subject row carries the observer's flag-free counts (space, wavenumber, time, frequency); the Carrier row the unitful measures (momentum, energy) anchored at the substrate, the crossing between the rows carried by the Planck constant: the $\{\hbar, h\}$ pair of §2.1. The continuum π below is always a labelled chart constant of such a drawing, never a claim.

2.1 The minimal configuration $(13, 233)$

The instance is the minimal non-trivial pair, and minimality is a finite scan under the admissibility predicate of §2: on the Subject side $\kappa = 1$ gives $\mathsf{p} = 5$, its own Q_4 core with no rung (the trivial pair $(5, 41)$ passes every clause), and $\kappa = 2$ gives 9, composite, so $\mathsf{p} = 13$ is the least shell with a rung; on the Carrier side the primes $\mathsf{p}^2 < \Omega < 233$ with $\Omega \equiv 1 \pmod 4$ all fail a clause (173, 181, 197, 229 fail $S \equiv 2$ mod 4; 193, with $S = 48$, fails both S-clauses), leaving $\Omega = 233$ (`verify-233`). All six claims of this section are re-derived by integer computation in `check_s13.js` and in the suite `verify-233`.

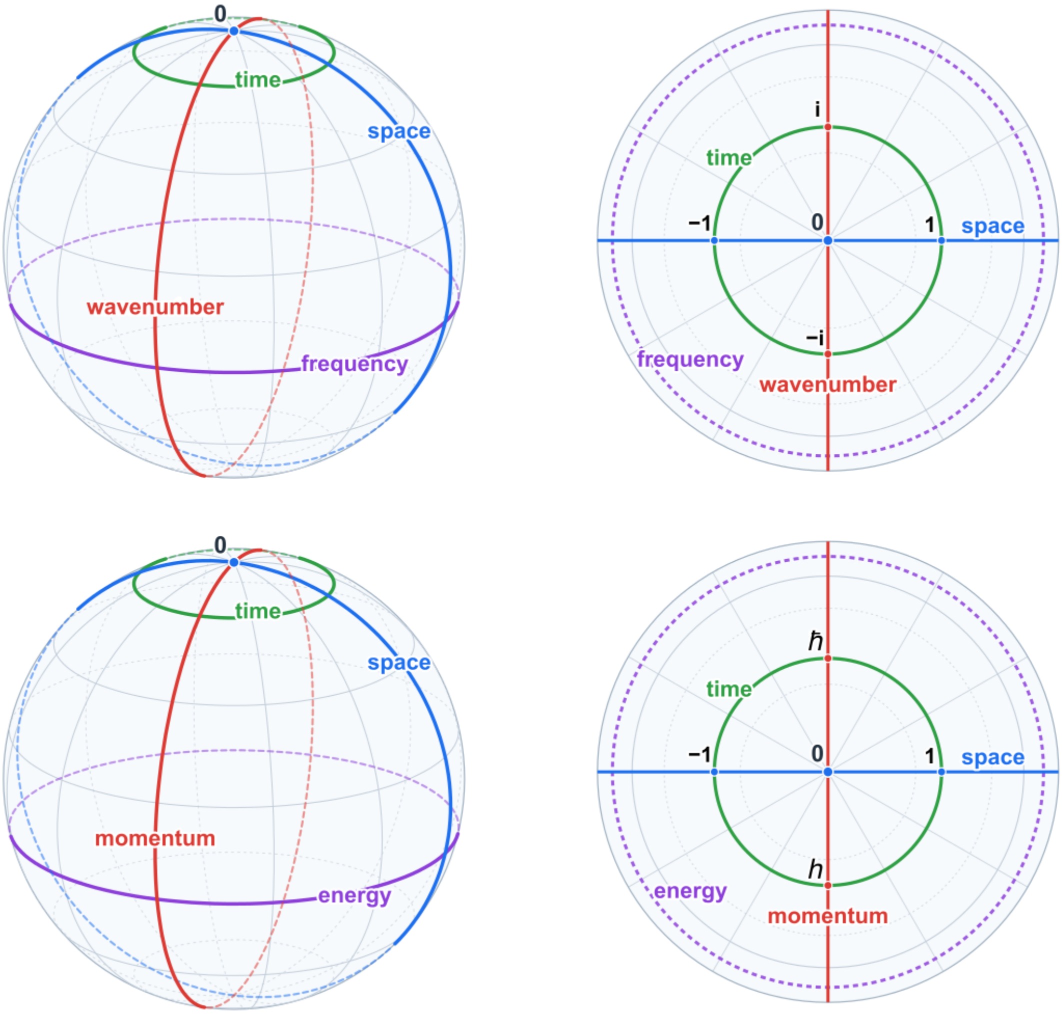


Figure 3: The four representation domains of a framed holographic shell, in both registers [33]. Subject register (top): space (the prime meridian, blue) and wavenumber (red), the transverse additive-Fourier pair; time (green) and frequency (purple), the longitudinal multiplicative-Fourier pair, the drive running along time; left the orbital sphere with the origin at the pole, right the observer's chart with the cardinal residues $0, \pm 1, \pm i$, the four counts flag-free and unitless relative to the frame's own anchors. Carrier register (bottom): the same domains with the dual charts named by their unitful readings, momentum and energy, anchored at the substrate; each dual chart differs from its count by exactly the quarter flag carried by the Planck constant; on the Carrier dial the quarter-turn role is held by $\hbar$ and the opposite role by $h = -\hbar$, with the quarter-root theorem drawn at its representatives: $(k_B c)^2 = (-2)(2^{-1}) = -1$ with no representative choice, so $k_B c$ is a quarter root, the choice of root representative data, and the drawn pair evaluates it as $124 \cdot 159 \equiv 144 = \hbar$.

A1: the constants. $\mathsf{p} = 13 = 4\kappa + 1$ with $\kappa = 3$; $\Omega = 233 = 4S + 1$ with $S = 58$. The drive $\mathsf{g} = 2$ generates $\mathbb{F}_{13}^{\times}$ (order 12); the quarter is $i = 5$, $i^2 = 25 = -1$; and, with heights read modulo $\mathsf{p} - 1 = 12$,

$$i \;=\; \mathsf{g}^9 \;=\; \mathsf{g}^{-3} \;=\; \mathsf{g}^{-\kappa}, \qquad -\mathsf{g}^3 = -8 = 5 = i.$$

A2: the tower identity. The drive cycle factors as $C_{12} = C_4 \times C_3$: the quarter $\hat{I} = \mathsf{g}^{-\kappa} = 5$ (order 4) and the rung ladder $\times 3 = \mathsf{g}^4$ (order $\kappa = 3$). The heights $4 - 3 = 1$ compose to the drive itself:

$$\mathsf{g} \;=\; \mathsf{g}^4 \cdot \mathsf{g}^{-3} \;=\; 3 \cdot 5 \;=\; 15 \;=\; 2 \quad \text{on } \mathbb{F}_{13}:$$

one chronon is one scale step times one quarter turn. The exponents are units exactly at $\kappa = 3$; the factorization is the channel of Theorem 6 acting inside the Subject. The general form is a theorem: one chronon is a rungs composed with b quarter turns whenever $4a - \kappa b \equiv 1 \pmod{4\kappa}$, solvable for every odd κ; $(a, b) = (1, 1)$ exactly at $\kappa = 3$, with $(4, 3)$ at $\kappa = 5$ and $(2, 1)$ at $\kappa = 7$ (`check_s13.js`).

A3: the channel at the instance. $\gcd(\mathsf{p} - 1, \mathsf{p} + 1) = \gcd(12, 14) = 2$: the winding sector meets the boost base in the sign alone. $\gcd(\mathsf{p} - 1, 2(\mathsf{p} + 1)) = \gcd(12, 28) = 4$: it meets the spinor cover in exactly Q_4. The quarter is spinorial, $N(i) = i^2 = -1$ (Theorem 6).

A4, A5: the Carrier quarter roots. On $\mathbb{F}_{233}$, with the Carrier chart generator 78,

$$78^{58} = 89 = h, \qquad h^2 = -1, \qquad \hbar = 144 = h^{-1} = -h, \qquad \hbar h = 1, \qquad \hbar + h = 233 = \Omega.$$

$\hbar$ is the i-oriented root, h the $(-i)$-oriented root [34]. The names $\{\hbar, h\}$, and the constant loci of Figure 4, are the registration reading of [30, 33]: canonical, Ω-uniform positions in the Carrier register, named by law and evaluated at the instance. The naming is consumed exactly once, by the registration orientation $\iota : i \mapsto \hbar$ (Proposition 7), and displayed in the charts; every proof consumes only the arithmetic of Theorem 3, $q^2 = -1$ and $q + q^{-1} = 0$.[1] The chart generator is declared: $78 = 3^{-1}$, the reframing gauge. The ring's four cardinals are the quarter operators $\{1, h, -1, \hbar\}$, and the octant $\zeta_8 = 97$ ($\zeta_8^2 = h$) sits at chart exponent $S/2 = 29$, carrying the odd crossing classes.

A6: the FRC algebra half. $\frac{1}{2} = (\mathsf{p} + 1)/2 = 7$ and $-\frac{1}{2} = 6$ on $\mathbb{F}_{13}$: the half seam of every cycle sits between rows 6 and 7, and 13/2 is not a chronon. Fractions below (13/2, 13/4, 13/5) are chart coordinates on drawn curves, never register values [29].

A7: the frame counts. $|\mathrm{PGL}_2(\mathbb{F}_{13})| = 13 \cdot 12 \cdot 14 = 2184 = 156 \times 14$: the cone chart carries $\mathsf{p}(\mathsf{p} - 1) = 156$ cells and the Hopf fibration 156 fibres of $\mathsf{p} + 1 = 14$ (Theorem 5).

The two registers at the instance. Figure 4 closes the primer at the paper's own pair: the shell a frame derives internally, beside the presented minimal Carrier. The Carrier ring displays the residue pairs whose unit-face magnitudes the observer registers as the physical constants; the root pair of -1 is $\{89, 144\}$, exactly the $\{\hbar, h\}$ of §2.1 (A4, A5), and the instance arithmetic of §2.1 reads every mark.

3 The laboratory, in detail

The laboratory of §1.1, panel by panel: no dependencies beyond a browser, every state exact and discrete, the screen projection a labelled chart approximation. Figure 5 shows the three panels at one state; Figure 6 the sky's three representations.

[1] The instance also registers the coincidences $13 = F_7$, $89 = F_{11}$, $144 = F_{12}$, $233 = F_{13}$ (Fibonacci indices) and $233 = 13^2 + 8^2$: instance-particular data, recorded here and consumed by nothing.

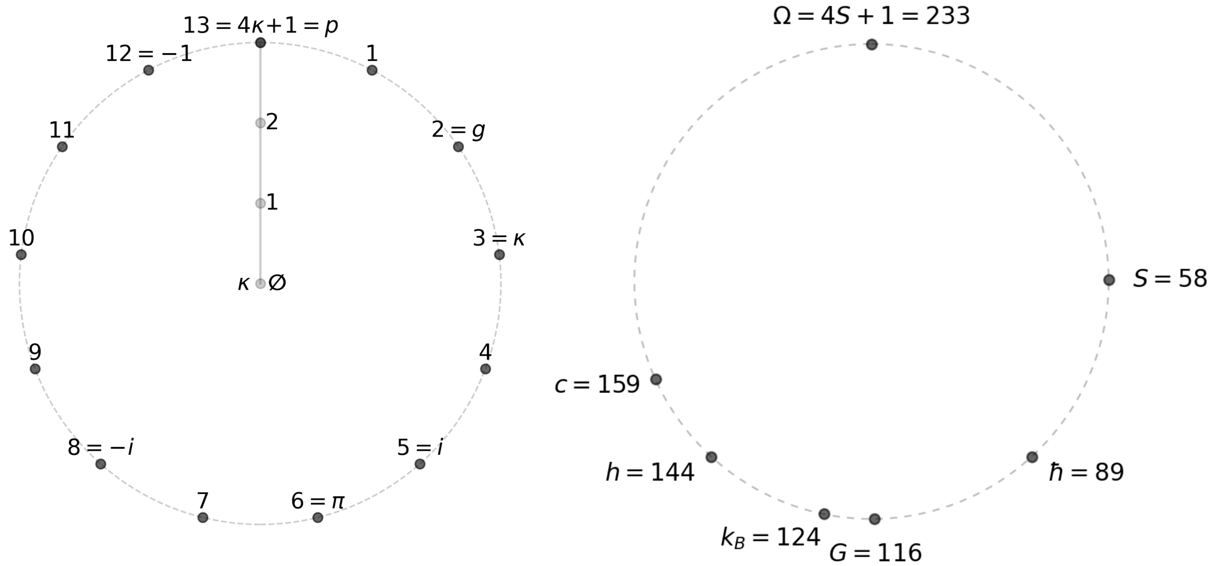


Figure 4: The two registers at the smallest admissible scale, $(\mathsf{p}, \Omega) = (13, 233)$ [33]. Subject (left): the shell $\mathbb{F}_{13}$ a frame derives internally (generator $\mathsf{g} = 2$, capacity $\kappa = 3$, quarter-turn $i = 5$, half-period $\pi = 6$, with $-i = 8$, $-1 = 12$, and the wrap point $\mathsf{p} \equiv 0$ carrying its framed label); the radial axis holds the shell's information capacity. Carrier (right): $\Omega = 233$, capacity $S = 58$, displaying the residue pairs whose unit-face magnitudes the observer registers as the physical constants: $G = 2S = 116$ exact at the half-cycle; the root pair of -1 is $\{89, 144\}$, the $\{\hbar, h\}$ pair, both members marked; c and k_B shown by the representatives 159 ($c^2 \equiv 2^{-1}$) and 124 ($k_B^2 \equiv -2$), pair partners 74 and 109. Marks are representative-convention data; the pairs are the Carrier facts: the flags are the content, the magnitudes their representative face, the physical-magnitude reading living in [33].

The holographic shell (Fig. 5b). The $\mathbb{F}_{13}$ arithmetic symmetry sphere: the graticule of meridians and latitudes, the rotating frame, and the register comb on the frame's great circle, each cell showing its whole value through the phase arrow i^r and the rung tint s: the fold $x = i^r \mathsf{g}^s$ of §2, drawn per cell. The node at longitude m, depth a carries the residue $a \cdot \mathsf{g}^m$: the meridians are the additive rays, the latitudes the drive orbits, every mark frame data. The arrow lies in the transverse plane of its meridian fiber, real axis radial and imaginary axis zonal, so one drive step turns every arrow one quarter about its fiber: the comb corkscrews around the space-like circle, the transverse-wave picture of the continuum recovered exactly. Four frame elements carry the four domains of Figure 3: the prime meridian (blue, space) and the quarter meridian (red, momentum), the transverse pair; the L_1 latitude (green, time), carrying the twelve register slots as its comb, and the L_4 latitude (purple, energy), the longitudinal pair. The Observer sits at the pole; the whole frame sweeps 30° per chronon in the Carrier view and stands in the Subject view.

The phase chart (Fig. 5c). The Carrier ring of 232 ticks, the standing backdrop of both views, with the two quarter roots at its poles: $\hbar = 144$ on top, i-oriented, and $h = 89$ below, $(-i)$-oriented, $\hbar + h = \Omega$ (A4); the octant marks $\zeta_8 = 97$ with $\zeta_8^2 = h$. Inside it the Subject ring C_{12}, its twelve slots with their phase arrows, the quarter at $i = 5$ and $-i = 8$; the fold clock $r = \tau \bmod 4$ with the unit's lift as its hand; and the q/p dial. The dial carries two counters, the precession quarter and the perihelion tick: the two faces of the leak (C7), counted live as the Subject ring precesses against the Carrier count at the halved tick (B7). The dial hand rides q to h at $\tau = S$: the quarter station, the origin's twist exactly $\pi/2$.

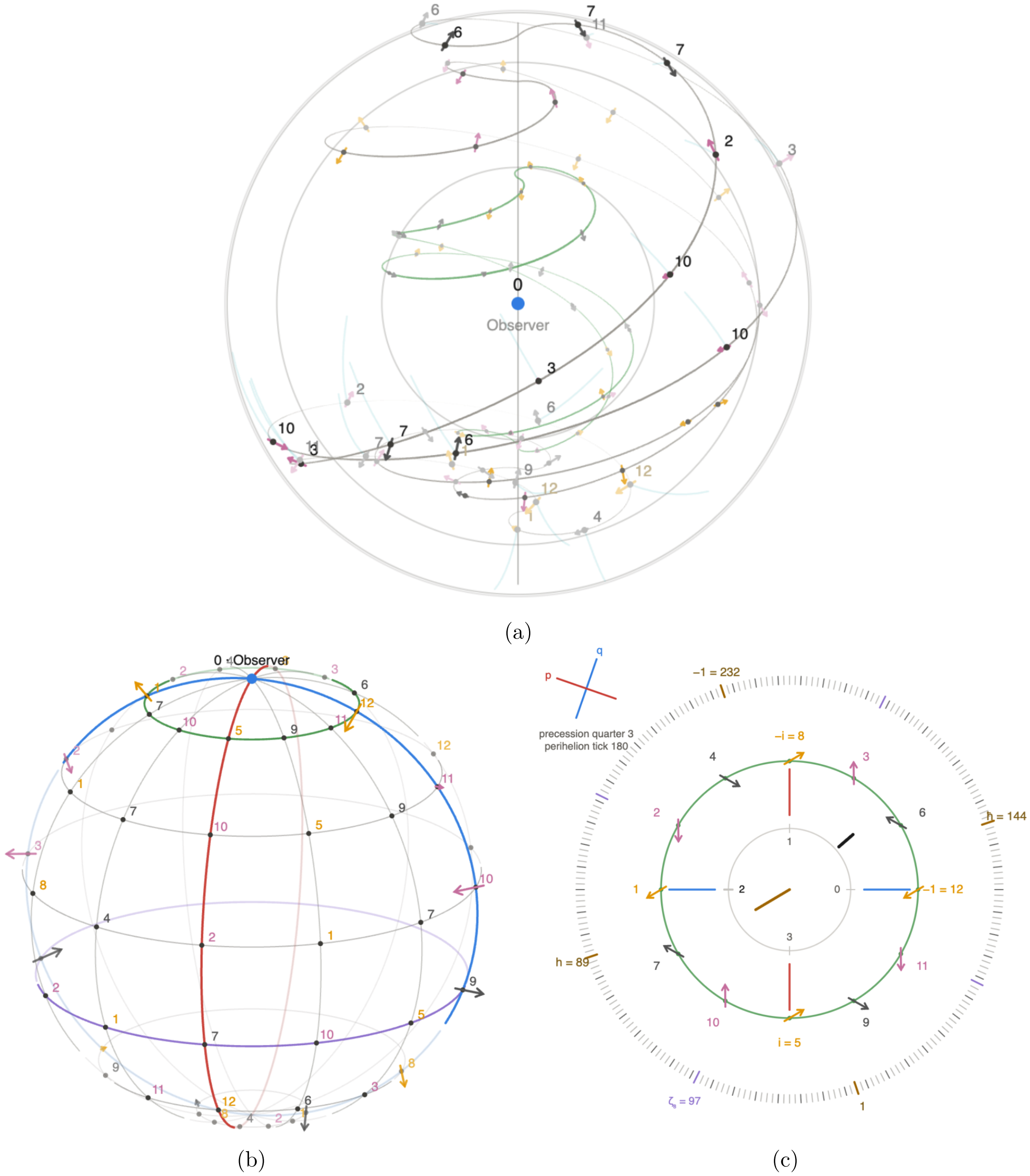


Figure 5: The three panels, one state, Subject view. (a) The sky: $\kappa = 3$ nested shells on the radial ladder, the Observer central, the 72-node field with each cell's phase arrow and rung tint (drive-orbit representation). (b) The holographic shell: the $\mathbb{F}_{13}$ sphere with its graticule, the frame's prime (blue) and quarter (red) meridians, the L_1 time latitude (green) carrying the register comb, the L_4 energy latitude (purple), the Observer at the pole. (c) The phase chart: the Carrier ring of 232 ticks with the action poles $\hbar = 144$ and $h = 89$ ($\hbar + h = \Omega$, A4) and the octant marks $\zeta_8 = 97$ ($\zeta_8^2 = h$); the Subject ring C_{12} with the quarter $i = 5$, $-i = 8$; the fold clock $r = \tau \bmod 4$; the q/p dial with the two leak counters, the precession quarter and the perihelion tick (C7).

The sky (Fig. 6). The three nested fibers of §6 with the representation toggle, the three factors of the frame group as three fiber layers over one node set. *Light cone* (Fig. 6a), the space factor: the M_0/M_3 pair, four rays each per route, the retarded ray and the advanced ray (the backwards-in-time shadow, the advanced cone $\times \mathtt{g}^{-1}$), every ray anchored at the drawn Observer and terminating at the quarter fold; the strokes are styled by projection depth alone, the front of the ball bright and the back faint, identically for all four rays; the pair is the drawn representative of the winding family, each meridian one label of the twelve under the one law, the full covering pinned arithmetically (Theorem 13). *Drive orbits* (Fig. 6b), the time factor: the helices of $x \mapsto \mathtt{g}x$, the retarded stitching of the time histories, with the time-factor loop highlighted. *Hopf fibers* (Fig. 6c), the boost factor: each node's exact leaf through its S^3 lift, the fibers filling the ball, mutually linked. The three representations share one sky: same node computation, same marks, same labels; the toggle switches the fibers only (C2). The Hopf mode draws the guard projection (centre $w_0 = 1.1$ outside S^3 [chart]) mapping the whole S^3 onto the ball whose silhouette is the horizon sphere; the chart is two-sheeted and the node's spinor sheet selects the preimage; the rung ladder is the foliation depth: Reading 10 made visible; motion along a fiber is a boost, changes the observer, and is unobservable; the nodes are the section (Reading 9).

The views and controls. The Carrier view shows the gauge: the common fiber flow, harness only, stepping clockwise with the drive. The Subject view shows the observable: the static relational curl of C1, Carrier-tick invariant. Pause, single step, and reset complete the controls.

The exactness boundary. Station incidence, causal ordering, winding, anchoring, and the registered flow are exact. The connecting curves are the declared spacetime chart: the guard projection (silhouette 206, chart scale $206\sqrt{0.21}$ [approx 94.40], seam [approx 85.82], tangency image 206), the $|\sin|$ radial interpolation, the tent law, the sheet-fair fold, the smootherstep ownership ramp — each declared [approx]. Certified: worst drawn corner 23° over all 232 Carrier ticks.

4 The structural theorems

This section carries the quantum sector's load-bearing structure at general $\mathsf{p} = 4\kappa + 1$, $\Omega = 4S + 1$. Proofs are elementary; each theorem names its machine witness at the instance.

Theorem 1 (Integration is quotient plus cover). *Let κ be odd and $S \equiv 2 \pmod 4$. Then $v_2(\mathsf{p} - 1) = 2$ and $v_2(\Omega - 1) = 3$; the shell and the Carrier share the class quotient C_4, and the shell closes on the double cover, the missing octant bit supplied as the second inversion. For κ even the shared quotient is C_8 and there is no cover. In both cases shared quotient times cover ratio is 8.*

Proof. $v_2(\mathsf{p} - 1) = v_2(4\kappa) = 2 + v_2(\kappa)$, which is 2 iff κ is odd; $v_2(\Omega - 1) = v_2(4S) = 2 + v_2(S) = 3$ for $S \equiv 2 \pmod 4$. The shared 2-part of the two cyclic groups is $C_{2^{\min(v_2(\mathsf{p}-1),\,3)}}$: C_4 for κ odd (cover ratio $8/4 = 2$, the double cover), C_8 for κ even (ratio 1). Witness: `verify-233` (the octant bridge block); instance $v_2(12) = 2$, $v_2(232) = 3$. □

Remark (the octant transport at the instance, after [31]): the shared quotient is transported through the coefficient plane $K = \mathbb{F}_{169}$ of the zonal theorem: $\gcd(168, 232) = 8$, an octant-equivariant counting carries the Fourier flip exactly, and the remainder $8(S/2 - \kappa(2\kappa + 1))$ cells close as whole octant fibres.

Theorem 2 (The spinor dichotomy). *The following are equivalent: (i) κ is odd; (ii) the drive folds as a product, $C_{\mathsf{p}-1} \cong C_4 \times C_\kappa$; (iii) $v_2(\mathsf{p} - 1) = 2$; (iv) the shell closes on the double cover with -1 at the half; (v) the fold congruence $3\kappa r + 4s \equiv 1 \pmod{4\kappa}$ is solvable.*

Proof. (i)⇔(iii) as above. (i)⇔(ii): $\gcd(4,\kappa) = 1$ iff κ odd, and then $C_{4\kappa} \cong C_4 \times C_\kappa$ by CRT. (iii)⇔(iv) is Theorem 1. (i)⇔(v): the values $3\kappa r + 4s$ modulo 4κ form the subgroup generated by $\gcd(3\kappa, 4)$, which contains 1 iff 3κ is odd iff κ is odd. Witness: `verify-233`; the instance is (i) with $\kappa = 3$. ☐

Theorem 3 (The Carrier quarter roots)**.** *Let* g_Ω *generate* $\mathbb{F}_\Omega^\times$ *and* $q = \mathsf{g}_\Omega^{(\Omega-1)/4}$*. Then* $\mathrm{ord}(q) = 4$*,* $q^2 = -1$*, and the two square roots of* -1 *are the inverse pair* $\{q, q^{-1}\}$ *with* $q + q^{-1} \equiv 0$*; the two roots are the two orientations of the quarter.*

Proof. $q^2 = \mathsf{g}_\Omega^{(\Omega-1)/2} = -1$ and $q^4 = 1$, so $\mathrm{ord}(q) = 4$ and the roots of -1 are $q^{\pm1} = \pm q$, summing to 0. Witness at the instance: $q = 78^{58} = 89 = h$, $\hbar = 144 = -h$, $\hbar + h = \Omega$ (`check_s13.js`). ☐

The Fourier reading of the quarter is imported, not proven here: on the representation space the scale action $x \mapsto qx$ is conjugate to the quarter-cycle finite fractional Fourier transform, carrying the additive-position chart onto the additive-momentum chart and back with a sign [34]. This paper consumes only the arithmetic of the theorem; wherever "Fourier flip" appears below it abbreviates the pair (exact quarter, imported reading).

Theorem 4 (The precession law)**.** *Run the pair one drive step per chronon: the shell on* $C_{\mathsf{p}-1}$*, the Carrier count on* $C_{\Omega-1}$*. Per shell revolution the shell's apse precesses against the Carrier quarter by the winding ratio: the apsidal fraction is exactly*

$$\frac{\mathsf{p}-1}{\Omega-1} = \frac{\kappa}{S},$$

for every admissible κ*-odd pair (*$S \equiv 2 \bmod 4$ *is consumed through Theorem 1). The joint cycle closes at* $\mathrm{lcm}(\mathsf{p}-1, \Omega-1)$ *at (home, home), and the half event at the exact midpoint is (home,* -1*).*

Proof. The apse advances $(\mathsf{p}-1)$ Carrier ticks per shell revolution against the Carrier cycle of $\Omega-1$ ticks; the fraction is $(\mathsf{p}-1)/(\Omega-1) = 4\kappa/4S = \kappa/S$. Let $L = \mathrm{lcm}(\mathsf{p}-1, \Omega-1)$ and $m = L/2$. By Theorem 1, $v_2(\Omega-1) = 3 > 2 = v_2(\mathsf{p}-1)$, so $v_2(m) = 2 = v_2(\mathsf{p}-1)$ and $(\mathsf{p}-1) \mid m$ while $m \equiv (\Omega-1)/2 \pmod{\Omega-1}$: at m the shell is home and the Carrier count reads -1. Witness: `verify-233` (precession block); instance $3/58$, $L = 696$, $m = 348 \equiv 0\ (12)$, $\equiv 116\ (232)$. ☐

Theorem 5 (The Hopf section)**.** *In* $\mathrm{PGL}_2(\mathbb{F}_\mathsf{p})$ *the non-split torus* $T \cong C_{\mathsf{p}+1}$ *acts on* $\mathbb{P}^1(\mathbb{F}_\mathsf{p})$ *without fixed points, and meets the Borel subgroup* B *(the stabiliser of a point, the affine group of order* $\mathsf{p}(\mathsf{p}-1)$*) trivially. Every frame factors uniquely as (cone-chart event)* $\times$ *(boost):* $G = BT$*, so* B *is a global section of the fibration* $G \to G/T$*, with* $\mathsf{p}(\mathsf{p}-1)$ *fibres of* $\mathsf{p}+1$ *elements. On the spin cover* SL_2 *the section obstructs at* -1*.*

Proof. A non-split element has irreducible characteristic polynomial over $\mathbb{F}_\mathsf{p}$, hence no eigenvector, hence no fixed point on $\mathbb{P}^1$; a fixed-point-free group meets every point stabiliser trivially, so $T \cap B = 1$. Then $|BT| = |B||T| = \mathsf{p}(\mathsf{p}-1)(\mathsf{p}+1) = |G|$ forces $G = BT$ with unique factorization, and B meets each coset gT exactly once: a global section. On the spin cover: in SL_2 the non-split torus is again cyclic of order $\mathsf{p}+1$ and contains $-I$, and the Borel and the torus now share the sign, $B \cap T = \{\pm I\}$: since $-I \in T$, the subset $BT \subset \mathrm{SL}_2$ is sign-saturated, of cardinality $\mathsf{p}(\mathsf{p}^2-1)/2 = |\mathrm{SL}_2|/2$, and it projects onto exactly half of PSL_2 (546 of 1092 at $\mathsf{p} = 13$; 30 of 60 at $\mathsf{p} = 5$). The section loses exactly one binary digit on the cover: the same one-bit deficit the construction reads throughout, as the octant bit of Theorem 1, the sign of the base channel of Theorem 6, and the halved tick of B7. The section cannot be globalised past -1: the spinor obstruction. Three objects are distinct and stay so throughout: the PGL_2 boost torus $C_{\mathsf{p}+1}$; its SL_2 counterpart, again $C_{\mathsf{p}+1}$, containing $-I$; and the norm-sign extension $C_{2(\mathsf{p}+1)} \subset \mathbb{F}_{\mathsf{p}^2}^\times$ of Theorem 6, which lives in the field, not in SL_2. Witness: `verify-hopf` (10 checks, at $\mathsf{p} = 13$ and, Ω-blind, at $\mathsf{p} = 5$). ☐

Theorem 6 (The mass–energy channel)**.** *Inside $\mathbb{F}_{\mathsf{p}^2}^{\times}$, the winding sector meets the boost base in the sign alone and the spinor cover in exactly the quarter:*

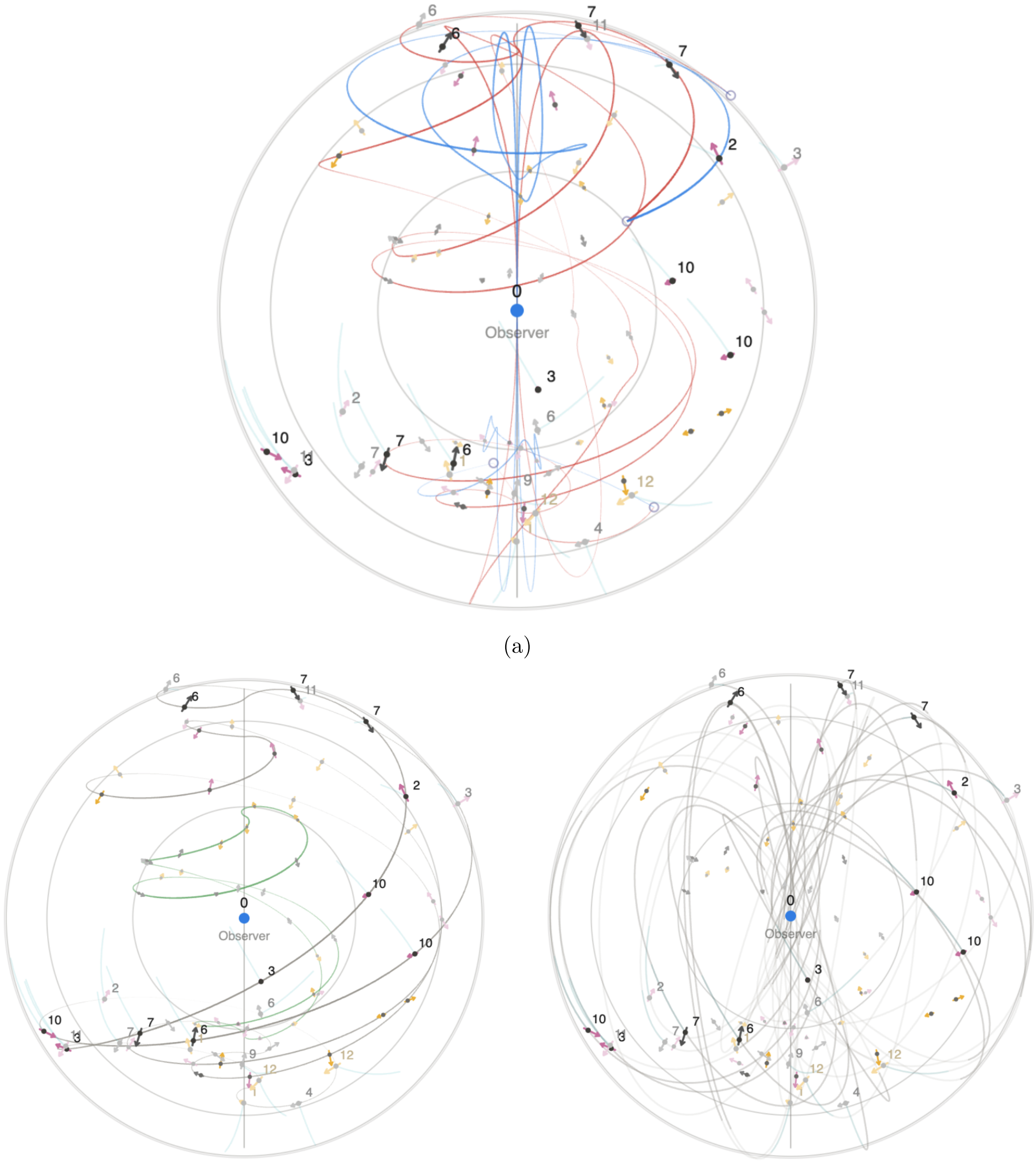
$$C_{\mathsf{p}-1} \cap C_{\mathsf{p}+1} = \{\pm 1\}, \qquad C_{\mathsf{p}-1} \cap C_{2(\mathsf{p}+1)} = Q_4,$$

for every $\mathsf{p} = 4\kappa + 1$. The quarter is spinorial: $i \in \mathbb{F}_{\mathsf{p}}$ is Frobenius-fixed, so $N(i) = i \cdot i^{\mathsf{p}} = i^2 = -1$, and the transport between the winding (energy) and norm-one (mass) channels crosses the sheet. Consequently the only rate datum transportable between the two channels is one Q_4 class; on the base, the sign alone.

Proof. $\mathbb{F}_{\mathsf{p}^2}^{\times}$ is cyclic of order $(\mathsf{p}-1)(\mathsf{p}+1)$; the intersection of its subgroups of orders a, b is the subgroup of order $\gcd(a, b)$. $\gcd(\mathsf{p}-1, \mathsf{p}+1) = 2$ for p odd. With $\mathsf{p} = 4\kappa+1$: $\gcd(4\kappa,\, 4(2\kappa+1)) = 4\gcd(\kappa, 2\kappa+1) = 4$, and $2(\mathsf{p}+1)$ divides $(\mathsf{p}-1)(\mathsf{p}+1)$ since $2 \mid \mathsf{p}-1$; so the second intersection is the unique $C_4 = Q_4$. The norm computation is displayed. Witness: `check_s13.js` regression-checks the proven gcd law to $\kappa = 10^5$ and the norm at $\mathsf{p} = 13$ ($i = 5$) and $\mathsf{p} = 53$ ($i = 23$); the kill test, the $\mathsf{p} = 53$ regression, is §7, C6. □

Proposition 7 (The register and its orientation)**.** *The frame $(\tau; 0, 1, \mathsf{g})$ is Subject-only, with shell data capacity-first: $\mathsf{p} = 4\kappa + 1$, $i = -\mathsf{g}^{\kappa}$, $\pi = 2\kappa$, $e = \mathsf{g}^{i}$, and the web closes on every shell: $2\pi \equiv -1$, $\mathsf{g}^{\pi} \equiv -1$, $-\pi = 2^{-1}$. The orientation is derived, not conventional: admissibility (S even) annihilates orientation transport, pullback covariance forces the phase $-s$ per chronon, and count positivity selects the material mode, fixing the oriented quarter $i = -\mathsf{g}^{\kappa}$; the joint flip $(\mathsf{g}, i, s) \mapsto (\mathsf{g}^{-1}, -i, -s)$ preserves every registered count and is the matter–antimatter gauge, selected factually by the instance's content. The Euler identity $e^{i\pi} \equiv -1$ holds exactly on the odd member of the $\pm\sqrt{-1}$ pair, and the parity names the gauge.*

Proof sketch. $i^S \in \{\pm 1\}$ for S even, so no orientation survives transport and the choice must be made frame-side; covariance of the pullback under reframing forces the mode phase $-s$ per chronon; positivity of registered counts picks $s = +1$; then the quarter through the material mode is $i = (\mathsf{g}^{-1})^{\kappa} = -\mathsf{g}^{\kappa}$. The identities are one-line exponent computations modulo $\mathsf{p} - 1$; $e^{i\pi} = \mathsf{g}^{2\kappa i^2} \equiv (-1)^i$, which is -1 exactly on the odd representative. Exponent arithmetic throughout is height arithmetic: an exponent is an integer lift reduced modulo $\mathsf{p} - 1$ (A1), so $e^{i\pi}$ is e raised to the height $i\pi = 5 \cdot 6 = 30 \equiv 6 \pmod{12}$, i.e. $6^6 = -1$; the residue product $i\pi$ mod p never enters an exponent. Full derivations: [30, 32]; instance witness: `check_s13.js`. □



(a)

(b)

(c)

Figure 6: The representation toggle: one sky, one state, one node set; the toggle switches the fibers only (C2). (a) Light cone: the observable meridian M_0 (blue, winding 1) and the momentum dual M_3 (red, winding 5), the retarded and the advanced rays per route, anchored at the Observer, terminating at the quarter folds; the strokes are styled by projection depth alone, front of the ball bright, back faint; the radius is the scale coordinate $R|\sin(2a\pi/13)|$. (b) Drive orbits: the helices of $x \mapsto \mathfrak{g}x$, the retarded stitching of the time histories, the time-factor loop highlighted. (c) Hopf fibers: each node's exact leaf through its S^3 lift, drawn by the guard chart as a bounded ellipse; the fibers fill the ball, mutually linked, the three shells at three foliation depths (Reading 10); the nodes are the section, motion along a fiber a boost, unobservable.

Realisation 8 (Time, mass, distance, quantum)**.** *The model's physical terms, each denoted in the substrate and tagged as such: constitutive realisations, not interpretive layers, each defended by its registered consequences.* ***Time is scale-dilation****: each Subject rides its own drive and assigns its own tick; a composite is read through the reader's one drive, so diagonal action and conserved offset are observer-derived; the Carrier contributes no clock.* ***Mass is winding rate*** *($E = hf$ an identity): a mass is the non-split (Frobenius) component the drive rotates; masslessness is split-torus eigen-alignment. The identity types within one register: h is the quarter flag between a count and its dual chart, not a magnitude, so the product never crosses moduli; the magnitude reading lives in [33].* ***Distance is decoherence****: the separation of two systems is their decoherence count in the observer's chart.* ***Quantum****: matter is the phase character of a finite cycle; observation is shared-core comparison; the complex amplitude is forced by Q_4 ($i^2 = -1$) [31, 32].*

Reading 9 (Observation is the Hopf section)**.** *The Subject observes its past light cone, retarded one chronon per space quantum. The cone chart is the affine group of Theorem 5: $\mathsf{p}(\mathsf{p}-1)$ cells, the Borel, a global section of the Hopf fibration; the boosts $C_{\mathsf{p}+1}$ change the observer; the horizon is the origin's antipode $\mathsf{p}/2$, the continuum ∞ [29].*

Reading 10 (The shell is holographic)**.** *By Theorem 5 every frame of the observable variety is one shell-trace event times one projective direction, 156×14 at the instance, nothing left over. The conjectured geometric reading: the variety is the combinatorial S^3; its finite 2-sphere is the quotient base of $\mathsf{p}(\mathsf{p}-1)$ cells — distinct from the $(\mathsf{p}+1)$-point projective direction set and from the drawn orbital sphere of the shell — the $C_{\mathsf{p}+1}$ cosets its Hopf circles, the shell its unique boundary chart.*

Statement 11 (The physical floors and the gravitational face)**.** *Two further readings the laboratory touches. (i) Ladder-floor objects are trivial shells: the smallest Object realisation is the bare proton ($\kappa_O = 1$); the first non-trivial one is ground-state hydrogen, $\kappa_O = 3$, forced, since $\kappa_O = 2$ has no shell. (ii) The gravitational face, in two parts, the first orbit-free.* The coefficient identity. *The pair carries two exact co-rates, the angular face κ/S (the apsidal fraction, Theorem 4) and the temporal face $\kappa/(2S)$ (the ring at the halved tick, §6 B7): ratio exactly 2, the double cover (Theorem 2), κ and S cancelling. A parametrised-post-Newtonian metric carries the same two faces with ratio $(2/3)(2 + 2\gamma - \beta)$, the gravitational radius and the separation cancelling likewise. Agreement is exactly $2\gamma - \beta = 1$, the deviation $(2/3)(2\gamma - \beta - 1)$: an identity between coefficients, consuming no orbit, no capacity numeral, and no registration; the exponential metric ($\beta = \gamma = 1$) satisfies it. The combination is a named discrimination, not only an abstract falsifier. In Brans–Dicke theory [35], $\beta = 1$ and $\gamma = (1 + \omega)/(2 + \omega)$, so $2\gamma - \beta - 1 = -2/(2 + \omega)$: nonzero at every finite coupling, with deviation $-4/(3(2 + \omega))$. The substrate forbids the scalar–tensor family outright; the combination returns to the forced value only in the $\omega \to \infty$ limit, where the theory is general relativity.* The registration dictionary. *Reading the temporal face as the circular-orbit clock deficit — the literal-dilation content of Realisation 8, no new premise — forces*

$$\tfrac{3}{2}\,\frac{r_g}{p_{\rm sl}} = \frac{\kappa}{2S} \implies p_{\rm sl} = 3\,(S/\kappa)\,r_g\,,$$

and the angular face then equals the metric's apsidal fraction $3\,r_g/p_{\rm sl} = \kappa/S$ automatically. At $\kappa = 3$ the dictionary reads $p_{\rm sl} = S\,r_g$: one gravitational radius per decoherence step, the separation count the capacity — the distance-is-decoherence reading of Realisation 8. The dictionary alone carries the scoping clauses: it is a one-point registration per (κ, S), the orbit- and mass-dependence of gravitation arriving as Object physics, and its continuum side is a first-order chart, the circular-orbit deficit at leading order; the coefficient identity above is independent of both clauses. Witnesses: `check_o1.js` *(exact) and* `o1_gr_chart.py` *(symbolic [approx: continuum comparison]). Within that scope the realisation stands with its falsifier: strong-field confirmation (§10).*

5 The register: the wave function is phase

The headline realisation of the model is the dictionary

$$i\hbar\,\frac{\partial\psi}{\partial t} = \hat{H}\psi \quad \rightleftharpoons \quad \psi_{\tau+1} = \mathtt{g}\cdot\psi_\tau \ :$$

ψ is the register itself, and each pure mode is a power map $u \mapsto u^k$ of the residues [31]. The step $\mathtt{g}\psi$ carries three exact readings: the permutation $P_{\mathtt{g}}|a\rangle = |\mathtt{g}a\rangle$ (the slots turn), the pointwise value walk (every registered value multiplies by $\mathtt{g}$), and the mode chart (phase $\mathtt{g}^{-k}$ per chronon on the winding-k character). The first two coincide because values are residue powers: multiplying a value by $\mathtt{g}$ is shifting a slot by one. By the tower identity (A2) the step reads

$$\psi_{\tau+1} = i\,(3\,\psi_\tau) \ :$$

one quarter turn times one scale shift per chronon.

The spectral structure is exact. The winding operator $H_{\text{wind}}\chi_k = k\,\chi_k$ generates the drive as $P_{\mathtt{g}} = \zeta_{12}^{-H_{\text{wind}}}$: the energy of a mode is its winding rate, the $E = hf$ identity of Realisation 8 read on the register (witness `verify-space`, 25 checks). Distinct from it is the Cayley kinetic operator $H_{\text{kin}} = 2I - T - T^{-1}$ of the shell dynamics, with $H^7 = 0$, $I - wH$ invertible, and the evolution U exactly unitary of order 13 (witness `verify-f13`, 25 checks). The two operators act on different objects and share only the sign sector: $P_{\mathtt{g}}$, of order 12, drives the register's units; U, of order 13, evolves the translation shell of the Cayley layer; free evolution is the first, interaction structure previews in the second, and no claim identifies them. An amplitude layer over the residues, with free coefficients, arrives only with an Object [32]; the present instance is Objectless, and ψ is phase throughout.

Proposition 12 (The resolvable spectrum)**.** *A winding $k \in \mathbb{Z}/(\mathsf{p}-1)$ is read at its symmetric height $\hat{k} \in \{-(2\kappa-1),\ldots,2\kappa\}$, and the capacity clause $4|\hat{k}| < \mathsf{p}$ — the station bound of D1, read on the winding cycle — selects the resolvable window $|\hat{k}| \le \kappa$. On the window the spectrum of H_{wind} is the integer interval $\{-\kappa,\ldots,\kappa\}$, bounded below by $-\kappa$; the registered energy is the height magnitude $|\hat{k}| \in \{0,\ldots,\kappa\}$, the sign the orientation datum. Pairwise conservation on the window is exact integer conservation: $|\hat{k}_1 + \hat{k}_2| \le 2\kappa$, strictly inside one period, no modular identification below capacity — up to orientation at the antipode $2\kappa \equiv -2\kappa$, the one class that is its own negative, where the saturating pair lands. Sums of three or more resolvable modes can wrap; multi-mode aggregation is Object physics, beyond this instance. The cyclic wrap is below resolution by construction. At the instance the window is $\{0,\pm1,\pm2,\pm3\}$: the drawn station set of D1 together with the ground level.*

Proof. Symmetric lifts order $\mathbb{Z}/(\mathsf{p}-1)$ on one period; $4|\hat{k}| < \mathsf{p} = 4\kappa+1 \Leftrightarrow |\hat{k}| \le \kappa$; for window heights $|\hat{k}_1 + \hat{k}_2| \le 2\kappa < 4\kappa = \mathsf{p}-1$, the bound attained only at the antipode. The station set is pinned in `check_s13.js` (D1). □

Three further identities complete the register doctrine. Read as a K-valued function on the cycle, the register comb is $\mathtt{g}^\tau \cdot \chi_1$, the shell's own fundamental character: the frame register is the fundamental winding, and Object modes are $k \neq 1$ selections read against it. The full space splits as $K\delta_0 \oplus K^{\mathbb{F}_{13}^\times}$, the step acting as the identity on the origin summand and as the drive D on the units: the register section is an instantaneous, space-like object, the residue vector of the frame's prime great circle. The active and the passive readings are duals through the inverse table: the drawn comb advances by D^{-1} while the coefficient wavefunction $\Psi_\tau = D^\tau\Psi_0$ advances by D, one dynamics read from two frames; and the register vector is isotropic, $\sum_a a^2 \equiv 0 \pmod{13}$, the zonal theorem's own isotropy of the winding lines [31].

6 The sky

The observable is drawn as a sky: $\kappa = 3$ nested shells on the radial ladder, the Observer at the centre. All claims are pinned by `verify-sky` (28 checks) and `verify-render` (20 checks).

B1: the node field. 72 nodes = 3 shells $\times 2$ routes $\times 12$ rows; the cell of node (e, k) is $c = e \cdot k \bmod 13$; the 24 outer labels are the full C_{24}. Labels are retarded: row k shows the value at $\tau - k$, i.e. $(e \cdot k \bmod 13) \cdot \mathsf{g}^{aS-k}$, one chronon of lookback per row.

B2: the radial ladder. Shells sit at $R \sin(2a\pi/13)$ [approx; chart]. The capacity bound $4a < \mathsf{p} \Leftrightarrow a \leq \kappa$ resolves exactly $\kappa = 3$ decoherence steps; the third shell grazes the horizon.

B3: the mounting. Consecutive shells are one rung apart, $\times 3 = \mathsf{g}^4$ (ladder $[3, 9, 1]$), mounted $120° = 8$ C_{24}-steps about the axis; the quarter is blind to both ($4 \equiv 8 \equiv 0 \bmod 4$): arrows repeat across shells, tints cycle one rung, labels divide by 3 inward. Each shell carries the complete aligned fiber, $48 = 4 \times 12$ cover-nodes; with the origin stabilizer, the pure drive, twelve null events, the count closes: $3 \times 48 + 12 = 156 = \mathsf{p}(\mathsf{p} - 1)$, the observable decomposed with nothing left over. Equal weight per shell is the uniform counting measure on the three C_3 fibres; the Born readout lives at the registration layer and consumes this measure [32].

B4: the sky chronon. $\tau \bmod 24$ is pair-native: the fibre product of ($t_S \bmod 12$) and ($t_C \bmod 8$) over their common C_4 ($\gcd(12, 8) = 4$, count $12 \cdot 8/4 = 24$, with $232 = 8 \cdot 29$) — the same shape as the registration fibre product of §2.1. The rendered sky closes at $t_{24} + 24$ and stands antipodal at $t_{24} + 12$: the double cover, drawn.

The sky map and its covers. The two chart angles exist through the quadratic extension $K = \mathbb{F}_{13}[\eta]$, $\eta = \sqrt{\mathsf{g}}$: the one further frame datum the observer supplies. The spinor clock $\langle \eta \rangle$ has order 24, the double cover of the drive cycle, with $\eta^{12} = -1$; every chart angle lands on the sky halved: the drive step 30° becomes 15°, the chart quarter 45°, the Carrier tick half a tick. Colatitude rides the spatial cover $C_{26} = 2\mathsf{p}$, the walk doubling the space circle: $\theta = k \cdot 180°/13$, the horizon falling strictly between rows 6 and 7, nothing registered on it. Azimuth is the sum of the two cover readings: the transport $(\tau + 3k) \cdot 15°$ on the spinor clock, direction-blind since decoherence is a count, plus the cell's own half-angle $c \cdot 180°/13$, the sign of -1 distributed one half-step per cell. The map is injective at every chronon: twenty-four images, no folds, both poles exact pass-throughs, one smooth closed loop. Per drive circuit the direct events number $12^2 + 12 = 156 = \mathsf{p}(\mathsf{p} - 1)$, one per element of the affine group: the cone chart of Reading 9 realized event by event; the two covers share exactly the one sign, $\gcd(24, 26) = 2$, and their central product glued over the common -1 carries $24 \cdot 26/2 = 312 = 2 \times 156$ cells (`verify-sky`). On the values the echo carries the conjugate relation: retarded $\times$ conjugate $= a^2$, the matter–antimatter gauge as the norm-circle relation (Proposition 7), the instance's matter content selecting the direct branch.

B5: the recurrence data. $696 = \operatorname{lcm}(12, 232) = S \cdot 12 = \kappa \cdot 232$, and the spinor recurrence $2 \cdot 696 = 1392$ on the Carrier cover C_{464}. Both are recurrence data of the declaring chart [proxy-chart], not pair ontology: the pair itself registers only the monotone dephasing.

B6: the Hopf representation. Each node lifts to its S^3 preimage and carries the exact Hopf leaf through it, node-on-fiber exact; the leaves rotate rigidly with the nodes (coherence residual 10^{-15} [chart]); the rung ladder is the foliation depth, $|z_2|$ levels [approx 0.14/0.69/0.86]: Reading 10, drawn.

B7: the frame leak. The Subject ring precesses against the Carrier count at the halved tick $\pi/232$; at the quarter station $b_C = S = 58$ the origin's twist is exactly $58 \cdot \pi/116 = \pi/2$; the wrap $\tau\, 695 \to 696$ is continuous, the half-angle branch carried by the sheet on C_{464} (Theorems 3, 4 at the instance). The frame ray moves at the composed rate $1/12 + 1/232$, the coprime numerator $\kappa + S = 61$: its absolute direction returns exactly at the pair event (home, home) and stands antipodal at (home, 116), the chart reading $\mathrm{dlog}(-1)$; between consecutive (home, home) events every class-compatible pair of cycle positions occurs exactly once, the pair space exhausted.

7 The flow

The dynamics drawn on the sky is a single retarded flow. Witnesses: `verify-233` (84 checks) and `verify-render`.

C1: one retarded flow law. Every registered row carries the flow of its emission chronon:

$$\varphi(k) = \Phi + k \cdot \mathrm{CURL} = -(b_C - k)\,\frac{\pi}{116}, \qquad \mathrm{CURL} = \frac{\pi}{116}\ [\text{chart unit}],$$

one Carrier tick-angle per chronon of lookback. The Subject view draws the gauge-free curl $k \cdot \mathrm{CURL}$ alone: static, integer-exact in the tick count, closing with the register (Theorem 4).

C2: one retarded flow across representations. The node images are computed by the same flow law in the drive-orbit, light-cone, and Hopf modes; the toggle switches the fibers, never the dynamics. Pinned by image-identity across the three modes at a deep-flow state. Its Dirac name is typed: the flow is exact and common; its reading as the Dirac flow resides with the quadrature and dictionary realisations (D10, D11).

C3: ownership. The flow dresses registered rows, i.e. other cells' registrations; the Subject's own locus is the frame origin and is never dressed: no chart displaces the origin of the observation. Drawn: every ray anchors at the Observer at every clock state (anchors $\leq 3 \times 10^{-12}$, station threading $\leq 4 \times 10^{-13}$ [chart residuals]), across parity and quarter states in both views.

C4: the half-turn is exact. The fiber rotation by π is the lift's antipode, and the sheet-fair chart is odd, $\mathrm{chart}(-q) = -\mathrm{chart}(q)$: the gauge's π-branch acts on the drawn ball as the exact point reflection, which fixes the Observer. Near the quarter states the swept residual passes the tangency circle: the rim kiss, the drawn spinorial quarter of Theorem 6.

C5: the C_4 stations. Row k reaches its congruent parity image $w_F = -w$ at $b_C = 116 + k$ (with $116 = (\Omega - 1)/2$) and is home at $b_C = k$; the parity station is congruent by the chart itself, gauge-free.

C6: the dilation instance and the kill test. The apsidal fraction per revolution is exactly $\kappa/S = 3/58$ (Theorem 4). The channel reading survives an Ω-blind arithmetic regression at $(\mathsf{p}, \kappa) = (53, 13)$, read in the auxiliary modulus $157 = 4 \cdot 39 + 1$ with generator 5 — not an admissible pair ($39 \equiv 3 \bmod 4$, $53^2 > 157$), and admissibility is not consumed by Theorem 6: the Object drive $v_O = 5^{13} = 22$ is *not* in the Subject cycle $\langle 5^3 \rangle$ ($3 \nmid 13$: the quantum index); the factorization $C_{12} = Q_4 \times C_3$ splits it into core quarter times outcome, $22^9 = 28 = \mathrm{gen}^{-39}$ and 22^4 of order 3; the idempotent $e \mapsto 117e$ projects the winding exponent, $117 \cdot 13 \equiv 117 \bmod 156$: mass phase and winding rate are one Q_4 datum read twice (Theorem 6); the apsidal ratio is $12/52 = 3/13 = \kappa_O/\kappa_S$, closure at $\mathrm{lcm}(12, 52) = 156$, half event at 78. Witness: `check_s13.js`, blocks (c)–(e).

C7: the two faces of the leak. The flow carries the leak twice, at two exact co-rates. The angular face is the curl (C1): one Carrier tick per chronon, κ/S per shell revolution — the apsidal fraction of Theorem 4. The temporal face is the ring (B7): the halved tick on the cover $C_{2(\Omega-1)} = C_{464}$, $\kappa/(2S)$ per revolution. The ratio is exactly 2, the double cover of Theorem 2: the channel datum of Theorem 6 read twice, full quarter on the cover, sign on the base — the half event at 348 is the Carrier -1. This pair of faces is the entire input of the gravitational dictionary (Statement 11). Witness: `check_o1.js` (10 checks, exact).

8 The light cone: the observable meridians

The space factor of the sky is the meridian family. Witnesses: `verify-render` and `check_s13.js` (station sets, covering, ramification, fusion).

D1: the capacity stations. $4d < 13$ with $d(k) = \min(k, 13-k)$ selects $k \in \{1, 2, 3, 10, 11, 12\}$: the observable meridian M_0 threads the direct nodes (cells $\pm1, \pm2, \pm3$ on L_1, L_2, L_3) and their echoes (cells $\mp1, \mp2, \mp3$ at rows $12, 11, 10$; colatitudes of direct and echo sum to π); threading residual zero (float 10^{-13} [chart]), both views, all 232 clock states; the rows are $k = 1, \ldots, 12$, and $k = 0$ is the origin itself, never a station (ownership, C3).

D2: the step law and the leak signature. The trace alternates sides by exactly

$$\frac{11\pi}{12} + \frac{\pi}{13} + \frac{\pi}{232}$$

per step: short of the half turn by exactly $\pi/156 - \pi/232$, the totality half-angle ($156 = \mathsf{p}(\mathsf{p}-1)$) against the halved Carrier tick — the frame leak written on the space trace. The echo azimuth $80 - 5a$ lands the echo nodes exactly.

D3: the scale-tower circuit. The meridian radius is $R\,|\sin(2a\pi/13)|$: $|\sin|$ of the tower angle, one full period per cycle, period points $a = 0,\ 13/2,\ 13$ (the FRC algebra half [29], between rows 6 and 7). The cycle closes because the tower is periodic: κ scale steps of four heights each, $4\kappa = \mathsf{p} - 1$, one drive revolution (A2).

D4: the photon travels scales (realisation). The radial coordinate is scale, not distance: approach to the origin is the decoherence descent $L_3 \to L_2 \to L_1$ into registration; absorption is the completed cascade; the continuation is down the scale-periodic tower. Nothing ever crosses the Observer's position; between the folds the circuit runs below resolution, unobservable and undrawn in the observable sector.

D5: four rays. The drawn observable sector per meridian: the retarded pair (per route: the absorption descent, and walked outward, the emission) and the advanced pair, the backwards-in-time shadows (the advanced cone $\times\mathsf{g}^{-1}$), each a fiber geodesic from the Observer to the observational horizon at the quarter fold $a = 13/4$. The four rays at the origin realise the four quarter classes: the Q_4 channel of Theorem 6, drawn.

D6: the momentum dual. M_3 is the same meridian law at winding $\hat{I} = 5 = \mathsf{g}^9 = -\mathsf{g}^3$ (the $\pm i$ line): stations at cells $\pm5t$, rows $5, 10, 11$ (direct) and $2, 3, 8$ (echo) on the tent shells, via the C_{26} double-cover fold with the route flip on odd half-winds; axis passages at $t = 13m/5$; the drawn rays contain $t = 13/5$ and $t = 52/5$ at radius $R\sin(2\pi/5)$ [chart].

D7: the covering. The winding family $m = 1..12$, both rays, six stations each ($144 = 2 \cdot 72$ slots) covers the node field completely: 72/72 nodes, 24 per shell; multiplicity 1 on odd rows and 3 on even rows — 36 nodes once, 36 thrice, $144 = 36 + 108$. Each drawn meridian is one label of the covering family; density in angle is the tower statement, capacity-bounded [approx: profinite refinement].

The covering, defined. A slot is a triple (m, ε, t): a winding $m \in \{1, \ldots, \mathsf{p}-1\}$, a ray $\varepsilon = \pm 1$, and a parameter $t \in \{1, \ldots, \kappa\} \cup \{\mathsf{p}-\kappa, \ldots, \mathsf{p}-1\}$. With $u = mt$ and the wrap count $w = \lfloor u/\mathsf{p} \rfloor$, the slot lands on row $k = u - \mathsf{p}w$ when w is even and $k = \mathsf{p}(w+1) - u$ when w is odd (the tent fold), on route $\varepsilon \cdot (-1)^w$ (the route flip on odd sheets), and on shell $\min(t, \mathsf{p}-t)$. A node (shell a, route e, row k) therefore has exactly four candidate slots, the windings $\pm M$ with $M = k\,a^{-1}$ at the parameters $\pm a$; matching the node's route forces the sheet-parity requirements (even, odd, odd, even) on the four in turn.

Lemma 1 (The all-shell parity table). *Fix a node (shell a, route e, row k), let $M = k\,a^{-1}$ and $A = \lfloor aM/\mathsf{p} \rfloor$, and write $\alpha = A \bmod 2$, $\bar{a} = a \bmod 2$, $\bar{M} = M \bmod 2$. Then $aM = \mathsf{p}A + k$ exactly, so $\bar{k} \equiv \bar{a}\bar{M} + \alpha \pmod 2$; the four candidates match precisely when $\alpha = 0$, $\alpha = \bar{a}$, $\alpha = \bar{M}$, $\alpha \equiv 1 + \bar{a} + \bar{M}$ respectively; and the eight-case table gives exactly three matches for $\bar{k} = 0$ and exactly one for $\bar{k} = 1$, at every shell, with the odd sector's single match at parameter a (direct) when $\bar{a} = 1$ and at $\mathsf{p} - a$ (echo) when $\bar{a} = 0$.*

Proof. $M = k\,a^{-1}$ gives $aM \equiv k \pmod{\mathsf{p}}$ with $0 < k < \mathsf{p}$, so $aM = \mathsf{p}A + k$, and $\bar{k} = \bar{a}\bar{M} + \alpha$ since p is odd. The candidates' wrap counts are A, $a - 1 - A$, $M - 1 - A$, $\mathsf{p} - a - M + A$ by the reflection identity applied four ways; imposing the required parities (even, odd, odd, even) yields the four displayed conditions. Substituting $\alpha = \bar{a}\bar{M} + \bar{k}$ and enumerating $(\bar{a}, \bar{M}) \in \{0,1\}^2$ for each $\bar{k}$ gives the counts and the positions case by case; the table is machine-checked against the constructed covering at $\mathsf{p} = 5, 13, 17, 29$ (`check_o2.js`). ☐

Theorem 13 (The covering parity theorem). *In the covering just defined, every node has multiplicity $\mu \in \{1, 3\}$ with $\mu = 3 \Leftrightarrow k$ even, and the odd sector's single slot is direct on the odd shells and echo on the even shells, at every shell of every admissible p (Lemma 1); in particular $\mu \geq 1$: the completeness of the covering is a corollary. The even sector carries the dual split, and over the whole cover the direct and echo slots balance exactly; at the instance the counts are 72/72 nodes, $144 = 36 + 108$, and the balance 72/72.*

Proof. Reflection identity: for $a, b \in \{1..\mathsf{p}-1\}$, $ab + (\mathsf{p}-a)b = \mathsf{p}b$ and the two residues sum to p, so $\lfloor ab/\mathsf{p} \rfloor + \lfloor (\mathsf{p}-a)b/\mathsf{p} \rfloor = b - 1$. A node (shell a, route e, row k) has exactly four candidate slots, the windings $\pm M$ ($M = k\,a^{-1}$) at parameters $\pm a$, with required sheet parities (even, odd, odd, even). By the identity four ways their wrap counts satisfy $A + B = a - 1$, $C + D = \mathsf{p} - 1 - a$, $A + C = M - 1$, $B + D = \mathsf{p} - 1 - M$, and the total $\mathsf{p} - 2$ is odd: an odd number of candidates match, so $\mu \in \{1, 3\}$ and no node is missed. The decider and the position law, at every shell, are Lemma 1; at shell 1 the deciding wrap specialises to $w(\mathsf{p}-k, \mathsf{p}-1) = \mathsf{p} - 1 - k$, whose parity is the row parity. Witness: `check_o2.js` (11 checks: the identities, the law, the position, the eight-case parity table, and the Ω-blind sweep at $\mathsf{p} = 5, 17, 29$). ☐

The grading is the base channel of Theorem 6 written on the resolved mesh: the sky is covered at average 2 (the double cover) and graded 2 ± 1 by the row's C_2 class. The even rows are the g-divisible rows; the odd rows carry the half-cell obstruction of D8 below and are reached exactly once.

D8: the half-cell ramification. All four M_3 rays pass one point exactly at $t = 13/10$ and $13 - 13/10$: cell 13/2, where the route degenerates ($6.5 = 13 - 6.5$), the tent degenerates, and the radius degenerates. General winding: $t_{1/2} = 13/(2m)$; M_0's ramification point is the origin itself ($r(13/2) = 0$, absorbed in the self-echo).

D9: the horizon fold fusions. M_0's retarded fold on route e *is* M_3's advanced fold on route $-e$, and conversely — exact at every clock state, from

$$5 \cdot 39 \equiv -13 \pmod{104} \ :$$

the quarter fold enacts the quarter, the duality exchange $E \leftrightarrow B$ with route and time both flipped; eight ray-ends fuse into four horizon points.

D10: the quadrature reading (realisation). M_0/M_3 are the two quadratures of one field $F = E + iB$, the packaging of the Dirac construction [31]: co-propagating at the unregistered origin — all eight rays launch tangent to the clock axis (pinned $\leq 0.03°$ [chart]), no transverse label existing there (ownership; isotropy) — with the internal quarter appearing off the origin (tangents cross $90°$ inside L_1) and the pair fusing at the horizon (D9).

9 Discussion

9.1 What the instance proves

Completeness. Every structural statement of §4 holds simultaneously at one pair, with nothing left over: the 43-row predicate ledger of §10.1 is the checklist, and each row is machine-pinned. The model is not a metaphor for quantum mechanics on a small set; it is the Objectless phase sector of the quantum dynamics, closed, at $\mathtt{p} = 13$.

Ω-blindness. No physical constant and no numeral of the physical capacity enters anywhere: the instance is the viability layer of the programme. Physical readings — the constants, the masses, the cosmological sector — live at the physical S and are claimed in the papers that derive them [30, 32], not here. What the instance proves is that the structure those readings consume exists, closes, and is drawable, in complete detail, at the smallest admissible scale.

Exactness. The boundary between integer-exact content and declared chart conventions is stated wherever it is crossed (§3); every [approx] token in this paper is a chart datum on which no claim rests. The discipline is checkable: the six suites and `check_s13.js` contain no floating-point comparison behind any exact claim.

The continuum contrast. An infinite substrate predicts, when nothing is added, a leak-free world. Exact embedding is always available there: infinite divisibility offers every subsystem a commensurable choice, a cycle that embeds exactly, closes exactly, and evolves unitarily with no deficit; nothing forces a gauge layer, a spinor doubling, or a quantum of action. The observed deformation package must then be postulated piecewise: the gauge principle imposed, the spin structure chosen, $\hbar$ and the couplings inserted as unexplained scales, $\beta = \gamma = 1$ measured. The finite pair forces the same package with no freedom, and derives its structure: the leak at one tick per chronon (B7, C1), the double cover with face ratio 2 (Theorem 2, C7), the Q_4 channel (Theorem 6), the PPN combination $2\gamma - \beta = 1$ (Statement 11). This asymmetry is an argument: a hypothesis that permits the observed structure while its rival forces it has failed to compress, and each unforced postulate is a bit it cannot explain. The comparison sharpens under interaction: the continuum is not leak-free in practice, since its embedding deficits return as ultraviolet divergences, infinite and removed by subtraction, where the finite substrate settles the same bill in one derived tick. The argument is evidential, not deductive: a continuum with the observed structure bolted on remains consistent. It also carries its falsifiable edge: the forced combination $2\gamma - \beta = 1$ (Statement 11), a coefficient identity carrying no free parameter and none of the dictionary's scoping clauses, not a tunable coupling. The count seals the comparison. At the programme level the physics rests on three realisations (matter is a phase character, time is dilation, a frame change is a gauge symmetry) and one imported number, the de Sitter capacity [23]; the present instance consumes the realisations alone, with no import and no free parameter. Textbook quantum mechanics rests on some six unmarked postulates, the Standard Model on nineteen measured parameters, general relativity on a separately assumed coupling: unmarked and underived within their own frameworks; a fitted parameter carries no kill criterion of its own.

9.2 The instance and physical reality

The standing of the pair. The pair $(13, 233)$ is the smallest admissible instance of the structure, selected for total verifiability; it carries no claim about the scale of the world. The physical reality the programme implies is the pair whose Carrier capacity is fixed empirically, by the observed de Sitter entropy of the cosmological horizon: $S \approx 10^{122}$ in Planck units, $\Omega = 4S + 1$ [23], a cardinality of order 10^{122}. That number is the programme's single quantitative import, and it is an observable: the size of the totality is read off the sky, not posited. Capacity counts cells, not microstate exponentials: S is the horizon count in Planck units, read directly as the scale of the register cardinality, the reading whose metrology, including the selection of the admissible integer within the empirical band, is developed in [36]; nothing in the present instance consumes it. One negation is worth stating explicitly, since the minimal instance invites the misreading: the proposal is not that reality is small; it is that reality is finite, with its cardinality an empirical datum.

Instantiation, not approximation. The relation of $(13, 233)$ to the physical pair is exact instantiation. Every structural theorem of §4 is proven for every admissible $\mathsf{p} = 4\kappa + 1$ and merely evaluated here at the smallest one; nothing is truncated, discretised, or approximated, so the same statements hold verbatim at capacity 10^{122}. The instance stands to the physical pair as one hydrogen atom stands to chemistry: complete in structure, minimal in scale. The Ω-blindness discipline of §9.1 is the transfer license: results that consume no numeral of S and no numeral of κ hold for every admissible pair by construction; a result consuming the numeral of κ transfers through its general form (A2).

Why smallness is a feature. At $(13, 233)$ the totality of the observable is enumerable: 156 events per drive circuit, 72 nodes, every claim checked to the last integer, nothing hidden in asymptotics. This is the finite model-existence reading of §1 made operational: consistency exhibited by explicit model, the laboratory the model-existence proof.

The continuum as the large-capacity look. What changes with capacity is resolution, not structure. The radial ladder densifies into the radial coordinate, the tick into apparent smoothness, the chart seams sink below resolution: the continuum is what a 10^{122}-cell register looks like from inside, the degenerate reading, never the substrate.

The hierarchy reading. The deformation rate offers one further reading, at the realisation layer. The leak runs at one tick per chronon, a fraction $1/(4S)$ of the cycle: at the instance $1/232$, visible on screen; at physical capacity, of order 10^{-122}, minute. On the dictionary of Statement 11 the weakness of gravity relative to quantum effects is this $1/S$ suppression of the frame leak: the same denominator that makes the world look continuous makes gravity look weak.

Scope. The instance is Objectless: ψ is phase, and amplitudes with their Born statistics arrive with Objects [32]. It holds one observer: time-varying dilation is relational between systems of different rates and likewise deferred. The boost sector is the declared sequel; the gravitational realisation stands with its strong-field falsifier open (§10). What the paper exhibits is that the quantum sector closes at every admissible scale, and closes checkably at the smallest one; what the world adds is one number, read off the sky.

Outlook. One open item is on record: the tower identity (A2) and the covering theorems (D7, Theorem 13) are candidates for promotion to the programme's master results, with the four-ray/duality structure (D5, D9) as a channel corollary. The winding family is pinned arithmetically (`check_o3.js`): all 144 covering slots lie in the drawn observable sector, drawn axis passages exist exactly for windings $m \geq 4$, and every sibling's ramification point $13/(2m)$ falls in the observable sector; the drawn pair M_0/M_3 is the family's representative. The boosts change the observer; their systematic study, the rank-3 projective frame chart, is the declared sequel.

10 Predictions and falsifiers

Each item is sharp, machine-tested, and falsifiable; the first four are internal to the instance and already pass, the fifth is the paper's one outward conjecture.

1. **The channel kill test.** The $\mathsf{p} = 53$ regression (auxiliary modulus 157): a single failed congruence among $22 = 5^{13} \notin \langle 5^3 \rangle$, $22^9 = 28$, $117 \cdot 13 \equiv 117$, $12/52 = 3/13$ kills the channel reading (Theorem 6). Passes: `check_s13.js`.
2. **The covering law.** Any node of the 72 covered $\neq 1$ (odd rows) or $\neq 3$ (even rows) times by the winding family kills the tower statement — now theorem-backed (Theorem 13): a violation would contradict the reflection identity itself. Passes: $144 = 36 + 108$ exactly, and Ω-blind at $\mathsf{p} = 5, 17, 29$.
3. **The fusion congruence.** $5 \cdot 39 \not\equiv -13 \pmod{104}$ would kill the duality exchange (D9). Passes at every clock state.
4. **The tower exactness.** The unit-exponent factorization $\mathsf{g} = \mathsf{g}^4\mathsf{g}^{-3}$ fails at any $\kappa \neq 3$: the identity is a property of the minimal spinor pair, not an artifact of choice (A2).
5. **The gravitational face.** The registration dictionary $p_{\rm sl} = 3(S/\kappa)\, r_g$ is derived (Statement 11); the face-ratio agreement on both sides tests exactly the combination $2\gamma - \beta = 1$. Falsifiers: a metric class with $2\gamma - \beta \neq 1$ admits no identification (the deviation is exactly $(2/3)(2\gamma - \beta - 1)$: `o1_gr_chart.py`) — in particular Brans–Dicke at every finite ω is excluded, $2\gamma - \beta - 1 = -2/(2+\omega)$ [35]; a κ-odd pair without the double-cover half event would break the pair side (none exists, Theorem 2); outward, a strong-field system whose apsidal fraction departs from κ/S under the dictionary kills the dilation realisation (a two-body registration: deferred to the Object layer with the orbit map).

10.1 The predicate ledger

The construction's dependency structure is collected here in one place, so that every result above can be traced to its witness and the surface a reader must accept is made explicit. One predicate per row, one status tag per row; the source column names the verifying suite or audit, and every T row is integer-pinned by it. The falsifiable content drawn from this structure is collected in §10. Rows whose witness is `verify-render` certify implementation invariants of the declared chart, the drawn laws; rows witnessed by the arithmetic audits certify theorems. The ledger is the paper's checklist: §2.1 carries block A, §6 block B, §7 block C, §8 block D.

tag	meaning
T	**Theorem.** Derived and machine-verified in exact arithmetic by the named witness.
D	**Definition.** A naming or declared chart convention, carrying no empirical content.
R	**Realisation.** The row states what a physical term denotes in the substrate: constitutive, not interpretive, defended by the totality of its registered consequences.

#	Description	Status	Source
A. The pair: the instance arithmetic			
A1	The pair: $\mathsf{p} = 13$, $\kappa = 3$, $\Omega = 233$, $S = 58$; $\mathsf{g} = 2$, $i = 5 = \mathsf{g}^{-\kappa}$.	T	`check_s13`
A2	The tower identity: $C_{12} = C_4 \times C_3$, $\mathsf{g} = 3 \cdot 5 = 2$; unit exponents exactly at $\kappa = 3$; general form $4a - \kappa b \equiv 1 \pmod{4\kappa}$, solvable for every odd κ (Theorem 6 inside the Subject).	T	`check_s13`
A3	The channel at the instance: $\gcd(12, 14) = 2$, $\gcd(12, 28) = 4$; $N(i) = -1$ (Theorem 6).	T	`check_s13`
A4	The Carrier quarter roots, arithmetic: $78^{58} = 89 = h$, $h^2 = -1$, $\hbar = 144$, $\hbar h = 1$, $\hbar + h = \Omega$; octant $\zeta_8 = 97$, $\zeta_8^2 = h$.	T	`check_s13`
A5	The naming $\{\hbar, h\}$ and the constant loci of Fig. 4: the registration reading of [30, 33]; consumed once, by the registration orientation $\iota : i \mapsto \hbar$ (Proposition 7); proofs consume A4 alone.	**R**	Fig. 4
A6	The FRC algebra half [29]: $\frac{1}{2} = 7$, $-\frac{1}{2} = 6$; the seam between rows 6 and 7.	T	`check_s13`
A7	The frame counts: $2184 = 156 \times 14$; SL_2 obstructs at -1 (Theorem 5).	T	`check_s13`
A8	The resolvable spectrum (Proposition 12): symmetric heights, window $\lvert\hat{k}\rvert \leq \kappa \Leftrightarrow 4\lvert\hat{k}\rvert < \mathsf{p}$, energy the magnitude with the sign as orientation, pairwise-exact conservation below capacity; instance window $\{0, \pm 1, \pm 2, \pm 3\}$.	T	`check_s13`
B. The sky: the drawn observable			
B1	The node field: $72 = 3 \times 2 \times 12$; retarded labels, one chronon of lookback per row.	T	`verify-sky`
B2	The radial ladder: shells at $R \sin(2a\pi/13)$; capacity $4a < \mathsf{p} \Leftrightarrow a \leq \kappa$.	T	`verify-sky`
B3	The mounting: $\times 3 = \mathsf{g}^4$, $120^\circ = 8$ steps, quarter-blind.	T	`verify-sky`
B4	The sky chronon: $\tau \bmod 24$, the fibre product of C_{12} and C_8 over C_4; antipodal at $+12$, closed at $+24$.	T	`verify-render`
B5	The recurrence data: $696 = S \cdot 12$, spinor 1392 [proxy-chart]; the pair registers only the monotone dephasing.	T	`verify-233`
B6	The Hopf representation: node-on-fiber exact; foliation depth = rung ladder (Reading 10).	T	`verify-render`
B7	The frame leak: halved tick $\pi/232$; $\pi/2$ at the quarter station; continuous wrap on C_{464} (Theorems 3, 4).	T	`verify-render`
C. The flow			
C1	One retarded flow law: $\varphi(k) = -(b_C - k)\pi/116$; the Subject draws the gauge-free curl (Theorem 4).	T	`verify-render`
C2	One retarded flow across the three representations (its Dirac reading with D10, D11); the toggle switches fibers only.	T	`verify-render`
C3	Ownership: the origin never dressed; rays anchor at the Observer at every clock state.	T	`verify-render`
C4	The half-turn exact: odd chart, point reflection; the rim kiss (Theorem 6).	T	`verify-render`
C5	The C_4 stations: parity image at $b_C = 116 + k$, home at $b_C = k$, gauge-free.	T	`verify-render`
C6	The dilation instance 3/58; the (53, 13) kill test (Theorems 4, 6).	T	`check_s13`
C7	The two faces of the leak: angular κ/S (the curl), temporal $\kappa/(2S)$ (the ring, on C_{464}); ratio exactly 2, the double cover.	T	`check_o1`
C8	The face-ratio identity, orbit-free (both sides' scales cancel): matching the two faces (C7) against the isotropic metric forces $2\gamma - \beta = 1$ and excludes Brans–Dicke at every finite ω ($2\gamma - \beta - 1 = -2/(2 + \omega)$); the continuum side is the declared comparison chart [approx: continuum chart].	T	`check_o1`, `o1_gr_chart`
C9	The gravitational realisation: the registration dictionary $p_{\mathrm{sl}} = 3(S/\kappa)r_g$ derived from C8 and the dilation realisation (Statement 11); a one-point registration per (κ, S), the orbit- and mass-dependence deferred to Object physics, its continuum side a first-order chart; the realisation stands with its falsifier: strong-field confirmation.	**R**	`check_o1`
D. The light cone: the meridians and the covering			
D1	The capacity stations: $4d < 13$ selects $\{1, 2, 3, 10, 11, 12\}$; threading exact, all 232 states.	T	`verify-render`
D2	The step law $\frac{11\pi}{12} + \frac{\pi}{13} + \frac{\pi}{232}$; the leak $\pi/156 - \pi/232$ on the space trace.	T	`check_s13`

#	Description	Status	Source
D3	The scale-tower circuit: radius $R\|\sin(2a\pi/13)\|$, period at the FRC algebra half.	T	`verify-render`
D4	The photon travels scales: absorption is the completed decoherence cascade; nothing crosses the Observer.	**R**	`verify-render`
D5	Four rays realise the four quarter classes (Theorem 6), each a fiber geodesic to the quarter fold $a = 13/4$.	T	`verify-render`
D6	The momentum dual M_3 at winding 5: stations, tent shells, C_{26} fold; framed-rational passages 13/5, 52/5.	T	`check_s13`
D7	The covering: 144 slots cover 72/72; multiplicity 1 odd / 3 even; $144 = 36 + 108$.	T	`check_s13`
D8	The half-cell ramification: $t_{1/2} = 13/(2m)$; M_0's at the origin.	T	`check_s13`
D9	The horizon fold fusions: $5 \cdot 39 \equiv -13$ (104); eight ends, four points; $E \leftrightarrow B$.	T	`check_s13`
D10	The quadrature reading: M_0/M_3 the two quadratures of $F = E + iB$ [31].	**R**	`verify-render`
D11	The headline realisation: $i\hbar\,\partial_t\psi = \hat{H}\psi \;\rightleftharpoons\; \psi_{\tau+1} = \mathtt{g}\psi_\tau = i(3\psi_\tau)$.	**R**	`verify-space`
D12	The covering parity theorem, closed at every shell (Theorem 13, Lemma 1): $\mu \in \{1, 3\}$ by the odd four-sum; decider the row parity and position by the eight-case table; completeness a corollary; the carrier the sheet parity: even rows g-divisible (thrice), odd rows the half-cell obstruction of D8 (once).	T	`check_o2`
D13	The family law: the one meridian law generalized (`baseMW`; M_0, M_3 its windings 1, 5), the covering arithmetic fully pinned; the drawn pair is the family's representative, a drawn ten-sibling layer retracted as washing the sky.	T	`check_o3`
V. Machine verification			
V1	Six suites, 192 checks, all passing (§10.2).	T	all
V2	The exactness boundary: exact incidence/ordering/winding/anchoring/flow; connecting curves the declared chart, each constant [approx].	D	`verify-render`
V3	The in-tree audit: `check_s13.js`, 57 checks, re-derives every arithmetic identity of blocks A–D and the registration fibre product.	T	`check_s13`
V4	The dictionary audit: `check_o1.js` (10, exact) and `o1_gr_chart.py` (10 identities, symbolic rationals [approx: continuum chart]).	T	`check_o1`
V5	The multiplicity audit: `check_o2.js`, 11 checks, exact, including the eight-case parity table of Lemma 1 at $\mathtt{p} = 5, 13, 17, 29$; the decider sweep Ω-blind at $\mathtt{p} = 5, 17, 29$, the home instance excluded from the blindness sweep by design.	T	`check_o2`
V6	The family audit: `check_o3.js`, 7 checks, exact (the winding-1/5 identity, the 144-slot containment, the covering multiset, the passages, the ramification).	T	`check_o3`

Collecting the tags separates the categories directly. *Imported* is no measured constant and no external dataset: the instance is Ω-blind, and the general theorems it instantiates are proven in §4; the two imported mathematical readings, the Fourier conjugacy of Theorem 3 and the continuum comparison chart of Statement 11, are tagged where they occur. *Realised* are A5, C9, D4, D10, and D11: the action naming $\{\hbar, h\}$ with its constant loci, the gravitational dictionary $p_{\rm sl} = 3(S/\kappa)\,r_g$, the photon travelling scales, the quadrature pair $F = E + iB$, and the Schrödinger dictionary $\psi_{\tau+1} = \mathtt{g}\psi_\tau$ — each stating what a physical term denotes, constitutive, defended by its registered consequences, and confronted through the falsifiers of §10. *Declared* is the single chart row V2, the exactness boundary. *Derived* is everything else: thirty-seven theorem rows, each pinned by the named suite or audit. The formerly open block is closed and re-homed: the face-ratio identity derived (C8) and the gravitational dictionary it carries realised (C9), the multiplicity grading proven and interpreted (D12), the family law pinned with the pair as its drawn representative (D13); the one queue item remaining, the master promotion of A2, D7, and D12, is corpus bookkeeping, not a claim of this paper.

10.2 Reproducibility

Every exact claim in this paper is machine-verified. The laboratory's six suites run under node with exit-nonzero-on-failure semantics: `verify-233` (84 checks: pair dynamics, events, octant bridge, precession, the mass–energy channel), `verify-sky` (28: the sky map, fibered covers, radial ladder, the central product isomorphism $\phi(a, b) = 13a + 12b \bmod 312$), `verify-space` (25: register, cone arithmetic, the spectrum H_{wind}, the curl algebra $\delta_k = 78^k$ with $\delta_{58} = h$), `verify-f13` (25: $H^7 = 0$, $\text{ord}(U) = 13$, U exactly unitary), `verify-hopf` (10: the finite Hopf fibration, Ω-blind at $\mathtt{p} = 5$), `verify-render` (20: the production rendering itself) — 192 checks, all passing. The in-tree audit `check_s13.js`, distributed with this paper's source archive under `validation/`, re-derives every arithmetic identity of §2.1–§8, and the registration fibre product of §2, by integer computation: 57 checks, all passing. The dictionary audit `check_o1.js` (10 checks, exact rational arithmetic) pins the two faces and the derivation of Statement 11; its metric-side twin `o1_gr_chart.py` (10 identities, symbolic rationals end to end, the continuum side declared as comparison chart) derives the apsidal and clock-deficit coefficients from the isotropic metric by exact series. The multiplicity audit `check_o2.js` (11 checks, exact) pins Theorem 13 and Lemma 1 with the Ω-blind sweep, and the family-layer audit `check_o3.js` (7 checks, exact) pins the drawn covering family of §3. `README.md` maps each script to its claims.
Validation suite: https://github.com/gamayos/frc-numerics/tree/main/38-s13
Laboratory public link: https://frc-lab.gamma.earth/38-s13
Laboratory sources: https://github.com/gamayos/frc-numerics/tree/main/docs/38-s13

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Data availability. All observational inputs are published values cited in the text; the verification scripts are available at the linked repository.

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